

THE BIG CAVE

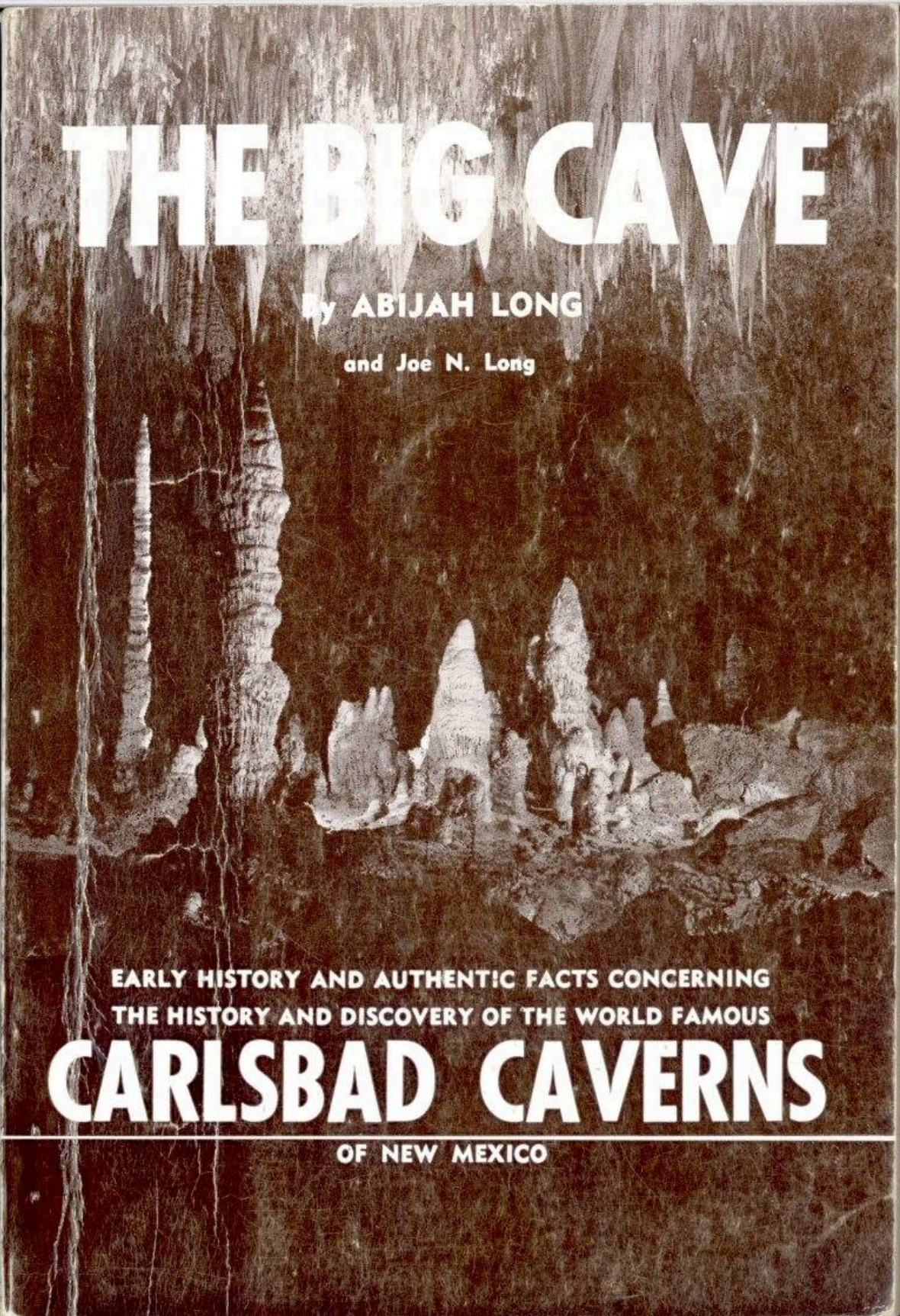
By ABIJAH LONG

and Joe N. Long

EARLY HISTORY AND AUTHENTIC FACTS CONCERNING
THE HISTORY AND DISCOVERY OF THE WORLD FAMOUS

CARLSBAD CAVERNS

OF NEW MEXICO



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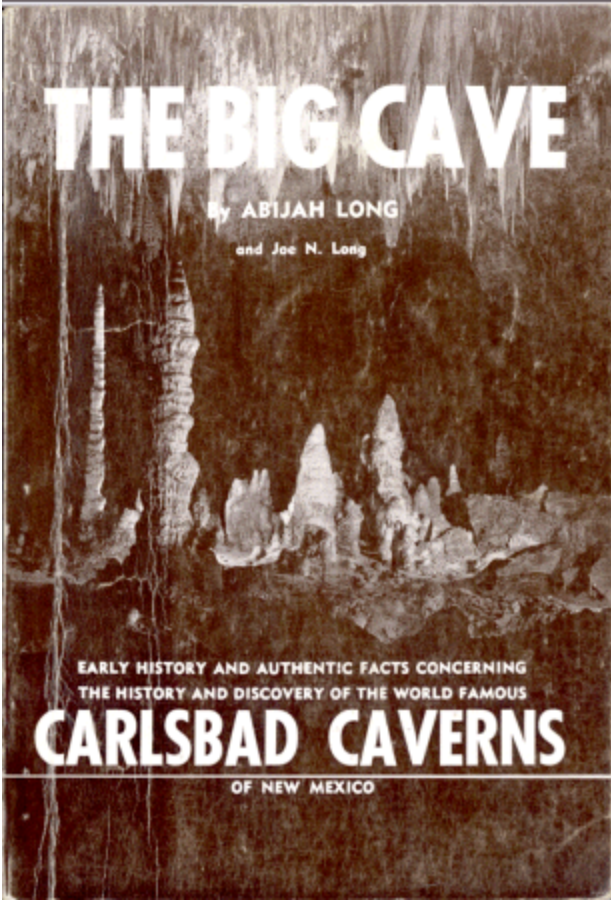
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*** START OF THE PROJECT GUTENBERG EBOOK THE BIG CAVE

The background of the book cover is a sepia-toned photograph of a cave interior. It shows several stalactites hanging from the ceiling and stalagmites rising from the floor. The lighting is dramatic, highlighting the textures of the rock formations.

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and

JOE N. LONG



Early History and Authentic Facts Concerning
the History and Discovery of the World
Famous Carlsbad Caverns of New Mexico.

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by

Mrs. Abijah Long, Joe N. Long, Mrs. Lou M. Wood,
Mrs. Kaye I. Williams, Ira B. Long, Mac A. Long.

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We Dedicate

*This Book to the Millions of Visitors who will
Follow our Father's Footsteps as They Come
to See and Enjoy the Mighty Wonders of the
Carlsbad Caverns.*

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[COVER PHOTO](#)—Totem Poles in the Big Room

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FOREWORD

(By the family of Abijah Long)

Prior to my husband's death in 1934, several of us had asked him to write, in his own words, the story of his early life, especially after moving to Carlsbad at the turn of the century.

He was reluctant to write of his experiences in connection with the Big Cave, as it was called in those days. But after considerable persuasion on our part he did sit down and write what happened in those early years following our move from Texas in 1901.

Since many historians today appear confused as to the actual beginnings of the Carlsbad Caverns, my children and I felt his words should no longer be for us alone, and we have therefore made them available in this form for all to read, and thus to know and understand more clearly just what happened during those early days of the cave's discovery.

My husband was always a very honest, though not always a prosperous, man. He was as good as his word, and in return he expected everyone else to be the same. At times, when he was forced to borrow money, he left only his word of honor and his promise to pay as collateral.

Nevertheless, we wanted to verify the facts in his story, and we have spent two years in tracking down many of the men who shared his experiences and in asking them to verify the incidents described.

We found many of them still alive and living in and near the charming community of Carlsbad. We found the people of Carlsbad friendly and courteous in every respect, and always eager to help us in our endeavor.

In gratitude for the kind assistance which met us everywhere, we wish to give our thanks to the friendly people of Carlsbad who so unselfishly assisted us in verifying my husband's account, especially the following:

Colonel Thomas Boles

"Dee" Harkey

Mrs. Simmonds of Happy Valley

Mr. and Mrs. John Queen
Johnny Forehand of Black River Valley
Mrs. Mary Queen Montgomery
Arthur Hoose
Wayne Crowder of Albuquerque.

We wish, also, to thank Robert Nymeyer, photographer of Carlsbad, who furnished the photographs of Carlsbad Caverns which we have used in the text.

Photographs of the lunchroom were furnished by the New Mexico State Tourist Bureau, Santa Fe, New Mexico, for which we are grateful.

We are also greatly indebted to various staff members of the National Parks Service who have made valuable suggestions to improve the technical accuracy of the manuscript.

It is our hope that persons interested in the Big Cave, now known the world over as the Carlsbad Caverns, will be enlightened as to the early history of the cave as told here for the first time by the cave's first owner, our husband and father, Abijah ("Bije"^[A]) Long, and its subsequent development as described by his eldest son, Joe N. Long.

Mrs. Abijah Long
Joe N. Long (Jodie in the story)
(Mrs.) Lou M. (Long) Wood
(Mrs.) Kaye I. (Long) Williams
Ira B. Long
Mac A. Long
(Mrs.) Anda M. (Long) Brubaker.

Carlsbad, New Mexico
November, 1956.

^[A] "Bije" was a nickname for Abijah. (Long "i" as in "hide.")

Part I
THE DISCOVERY

By ABIJAH LONG

1

We Move to Carlsbad

The distance from Goldthwaite, Texas, to Carlsbad, New Mexico, is slightly less than 400 miles—just a good day's drive in an automobile today. But in 1901 the automobile was something we heard about—something we read about, and friends of mine told of having seen a horseless carriage up in Dallas. People who did much traveling went by train or horse and wagon—or, they walked.

So when our family talked of moving West—talked of trying life anew "somewhere else," the question of how to go was considered. Train fare, we soon found, was much too high for us at the time. And with all our worldly possessions the freight charges would be excessive.

The answer—a covered wagon.

It was hard for my mother to leave Goldthwaite, and my wife was a bit fearful of the future in a strange land, but father and I thought our future was brighter if we tried anew somewhere else.

Besides my cousin, there were my two children, which completed our party of seven. Heading West over the vast expanse of open Texas prairie, I felt something of the thrill those early pioneers must have felt, although the fear of trouble was not present, for being ambushed by Indians was a remote possibility.

Just the same, there was an air of adventure in the journey, for none of us knew what the future held in store for us. We dreamed and hoped.

We completed our journey to Carlsbad, a distance of almost 400 miles, without any serious mishap. Oh, there were the usual little troubles of a sick horse which slowed us up one day, and an occasional steep hill which required we not only get out of the wagon to lighten it, but all help push as well.

Carlsbad had a population of less than a thousand in 1901, ^[B] and although Carlsbad was the legal name, having been officially changed two years before, everyone called the town by its original name, Eddy, and it was, and still is, the county seat of Eddy County.

^[B] Population in 1950, 18,000.

The primary activity in and around Carlsbad in those days was ranching, with a growing interest in mining of various kinds. I noticed many people continually coming and going and thought that, since I would have to get busy at something pretty soon, I would try the hotel business.

So, shortly after my arrival, I took over the Schlitz Hotel, which was located near the railroad station at the corner of Canyon Street and Mermod Avenue. The hotel boasted a dining room, and a large part of the business came from many of the railroad workers, who, it seemed, always had a good appetite and who liked good food and lots of it.

That wasn't the only trouble I had, and after a few months I found I didn't know enough about the hotel business to make a go of it, so I gave it up. The name of the hotel was later changed to Bates, and some time after that it caught fire and burned to the ground.

The saloon business in those days was always good. Ranchers and miners would come into town after a rugged week in the hills and spend much of their pay for liquor. It looked to me as though this would be a profitable enterprise and I decided to give it a try.

My wife didn't like the idea at all and told me so in no uncertain terms. However, I kept at it for a while in spite of her pleas.

One day I took my small son down to the saloon to show him off to the boys. That did it. That was the abrupt end to my venture in the saloon business. My wife set up such a howl that in order to keep peace in the family I decided to give up the saloon. Her opinion was always best for the family anyway, and of course I was aware of the fact that this was not the most desirable environment in which to raise children.

So, what next? I had by this time acquired a few mules and thought I might be able to use them in some way. I learned that the Joyce Pruitt Company of Carlsbad was expanding its operations and was in need of additional help in doing some freighting.

I made a deal with them to do the necessary hauling and soon was quite busy.

One of the biggest of my expenses was for food for the teams, and I was anxious to cut down the feed bill as much as possible. As a result, at the end of each day I would take the mules out to pasture and let them rest and graze.

At this particular time I had the teams grazing near a place known as Donahue Springs, now known as Oak Creek Springs. Water from the springs was a necessity for the mules, and we used the water ourselves, as well.

While the mules were grazing there wasn't much for me to do and I used to explore the countryside wondering what else might lie in the vicinity. Others came to Donahue Springs for water since in that arid country water was not as plentiful as we wished it might be.

One day a man by the name of Sam Evans and a Mr. Brown and myself were exploring the countryside just to see what we might discover in the area. Hunters and miners in that section were always telling of some unusual find and anyone who went out of town kept their eyes open in search of something that might turn out to be valuable.

I suppose that is why Brown, Evans and myself were searching—just looking around to see what we might uncover.

At one spot there appeared to be a hole or cavity in the ground and I called to my buddies, "Hey, come over here. There seems to be a hole in the earth."

In that rugged country a hole in the ground isn't exactly unusual, but this one seemed to lead to quite a large empty space beneath it, and the tone of my voice must have told the others it was not just an ordinary hole, because they both came over to where I was right away.

"What do you make of this?" I asked as they both gave the hole an appraising glance. The three of us were puzzled because it wasn't just a hole in the earth, but rather seemed to be the opening to a large cavity underneath.

"What do you suppose this is?" I asked again.

"Any large animals around here that might have dug it?" queried Brown.

"Possible," I reflected. "Let's have a look."

I was eager to go below and learn more about this mysterious opening in the earth, but it didn't seem to me that Evans or Brown shared my curiosity. We looked for animal tracks in the vicinity, but couldn't find any.

"Who'll go in with me?" I received no response. It didn't matter. By now I was so curious that I decided I would go in alone, if necessary.

I went to my wagon and got a lantern and a ball of large fishing line. Then I picked up all the rope I had, including the rope I used with my mules, and hurriedly went back to the mysterious hole.

I'll admit I was quite excited, and by now Evans and Brown were, too, although it seemed to me they tried to hide their enthusiasm. Perhaps they were just a bit afraid of what might be down there in that strange cavity, yet didn't want to let on that that was so.

I, too, decided to play it safe, so I took the lantern and tied it on to the end of the rope and lowered it into the hole. All three of us bent over the opening and looked in to see what we could see.

As the light of the lantern showed on the sides and bottom of the hole I could see that it was a large one. I particularly wanted to see if there was any damp in the cavity, but the light didn't reveal any.

By this time I was eager to go below and explore our discovery. "Who'll go with me?" I asked. No response.

"Well, then I'll go alone," I declared.

And with that I fastened the rope around my waist, made sure it was securely fastened to the ground a few feet outside the opening, and got ready to go below.

2

A Big Cave is Discovered

The three of us talked over the matter of my descent and several safety precautions were formulated. About this time I decided I would be much happier about the whole thing if I could persuade one of the others to go below with me. At least one should remain on the surface at all times in case anything should go wrong.

"Sam," I said, "how about you coming down with me?"

He seemed to want to see what would happen to me first, I guess, so I then asked him if I went first would he follow. He said nothing.

"We'll only stay down for a short while," I assured him. "Perhaps a half hour or so."

Sam did not like the idea of going down in such a place. I told him it was easy, so I made arrangements to go down into this cavity and I did.

After getting down in there I could look back out and talk to him, so I finally persuaded him to come down, which he did. That left Brown on top.

We prowled around in there for some time, exploring the cavity, being, as I remember, 75 or 100 feet deep.

When we got ready to climb out of our descension, Sam said to me: "How are we going to get out of here?"

eg. Sam did not like the idea of going down in such a hurry & told him that that was easy & made arrangements to go down into this cavity and a kid after me, getting down in there & could not get back and told me to run as I was, persuaded him to come down for which he did. But left Brown on top as we worked around in there for some time exploring & I said cavity that being old & remember 75 or 100 ft deep. When we got ready to climb out of our location Sam said to me how we were going to get out of here & told him that was easy he said go ahead & I said for go first and then I will come out. It is easy for me well Sam made a trial to go out but made a complete failure. Coming back down the rope in spite of me and told me we were in here for good & told me I could go out in less than 2 minutes he said he was not out & went. Sam would not try any more so I sent Mr. Brown to Carlsbad to get a rope long & rough that I might

A portion of the author's original manuscript.

I told him that was easy. He said, "Go ahead."

"No," I said. "You go first and then I will come out. It is easy for me."

Well, Sam made a trial to go out, but made a complete failure, coming back down the rope in spite of me, and said, "I guess we are in here for good."

I told him I could go out in less than two minutes.

He said, "Let's see you." So, out I went.

Sam would not try any more, so I sent Mr. Brown to Carlsbad to get a rope long enough that I might make a rope ladder out of it and lower it down to Sam.

Carlsbad was 28 miles away and, of course, travel was very slow with horse and wagon. I knew Brown wouldn't get back before the next day.

I called to Sam: "I'm sending Brown to Carlsbad to get enough rope for a ship's ladder. You'll be able to climb that all right and get out."

"How long do I have to stay down here?" was Sam's retort.

"Until he gets back with the rope," I answered. "Probably tomorrow."

"You mean I have to stay down here all night?" Sam's voice quivered with fear.

"Unless you have some idea of how to get out," I replied.

I brought my wagons and mules over to the hole and proceeded to get ready to spend the night there. In spite of Sam's anxiety he said he was hungry so I got some food from one of the wagons and lowered it down to him on the end of a piece of fishline. He still had the lantern with him. I don't know how much he relished his meal, but of course he had no choice; so, by the light of the lantern he ate his supper, such as it was.

Above, I fared somewhat better.

Sam was quite scared, and he remained squarely below the opening, which, by the way, was not over two feet in diameter. He had no desire whatsoever to take the lantern and go exploring by himself. I really do believe he meant it when he said he thought he was in there for good.

I didn't sleep much that night, and I don't think Sam slept at all. No matter how much I kept reassuring him that as soon as Brown returned from Carlsbad we would get him out, he still thought he'd never see the light of day again.

Early the next morning, before the sun was up, Sam was calling me and asking if Brown had returned. "Not yet," I replied, "but he will, probably some time before noon."

I, too, was getting a bit apprehensive, for I felt just a bit guilty about Sam's predicament, since I was the one who had persuaded him to go below—against his desire to do so. In order to slightly calm my nervous tension I decided to take a short walk around the area in hopes of making the time

pass more quickly, for I now knew that I, too, would welcome Brown and the rope he would be bringing.

I didn't stray too far from our campsite and was looking around when low in the sky I noticed a large number of dark objects. They seemed to disappear on a hillside.

Curiosity got the better of me. Knowing there was nothing I could do until Brown returned, I decided to investigate.

As I came closer I noticed the dark objects were bats—thousands of them. They were, indeed, disappearing into the side of a hill—into an opening that was much larger than the one which at the moment was holding Sam Evans a prisoner.

Having spent the night in search of food, they were now returning at sunrise to spend the day in what appeared to be a huge cave.

I was almost awe-struck at the sight of so many of these little flying mammals, for I had never before seen anything like it in my life.

As the brightness of the dawn increased the stream of bats subsided, and in a short while only a few remaining stragglers were entering the mouth of their home.

Where could all of these bats go, I wondered. There must be an exceptionally large cave inside to hold so many of them.

I never have cared much for bats, but I was interested in seeing where they lived. As I approached closer the whole side of the hill seemed to open up. There, certainly, was an opening to something even larger inside.

Well, of course, I wanted to go in, right then, but I knew it would not be safe to go into such a strange place alone. I had no idea of what I might find, or what trouble I might encounter.

By this time the sun was well off the horizon and I thought perhaps I had better get back to Sam and console him lest he think I had deserted him. I looked up the old dirt road towards Carlsbad to see if I could see any signs of Brown returning, but all was quiet.

I reasoned he would spend the night in Carlsbad and get an early morning start for the return trip, which would get him here a little before noon or so.

"Sam," I called out, "how is everything down there?"

"Get me out," he pleaded. "I've had all I want of this. I'm never going into a cave again."

I was quite excited about the larger cave I had just discovered and, of course, I wanted to tell Sam all about it, but when he said he never wanted to go into a cave again, I figured maybe he was in no mood to listen, let alone share my enthusiasm.

The best plan, it seemed, was to get his mind off his predicament so I tried to get him to talk about other things. He kept talking about how dark and cold it was down there in the hole, and when would Brown come back so that he could get out and why was he so foolish as to listen to me in the first place.

All this time I was eager to return to the new, big cave where all the bats were, yet I didn't want to leave Sam alone any more as I could see he was getting madder all the time. So the morning dragged on for him and for me as well.

Every few minutes he would call out to me and ask me to look and see if there was any sign of Brown. I had to keep saying no, but to say anything else would have raised false hopes, and I just couldn't do that.

About noon I noticed a cloud of dust on the horizon in the direction of Carlsbad, and as I watched it I could see it was getting bigger. That meant Brown was almost here. When I told Sam, he suddenly came to life again. I guess he felt like the condemned man who has just received a pardon.

Brown had gotten the rope, and we made a ship's ladder for Sam. We quickly lowered it through the small opening in the ground, and I never saw a man climb a rope ladder so fast before. Sam was mighty glad to see the light of day again.

Now that this ordeal was over I couldn't contain myself any more. I had to tell Sam and Brown of my discovery, and I wanted them to explore it with me. Certainly there was adventure ahead.

3

A Business Venture Appears

"Early this morning," I began, "while waiting for you to return, Brown, I took a walk around and noted a lot of bats going into a large cave,—a really large cave. Let's go have a look at it."

Sam didn't need to say a word. I could tell from his expression and his disappointment that he had no desire to go into any more caves, even if the entrance was on the side of a hill where he could safely get out.

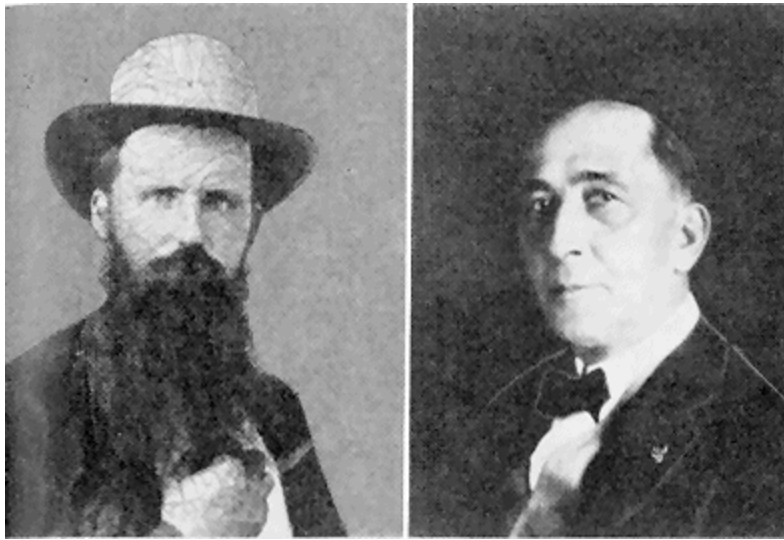
"Not me," he exclaimed in no uncertain terms. "Damn your cave. I'm going back to Carlsbad, and the quicker I get started, the better."

Sam would have it no other way, and since I felt partially responsible for his feeling the way he did, I decided to take him back to Carlsbad.

However, I couldn't get my mind off the new cave. I had to know what it was like inside. So, I made immediate arrangements to return.

I didn't want to go alone, so I inquired around and found two men who said they would be interested in going back with me and explore the cave.

The three of us set forth on the 28 mile journey to Donahue Springs, Andy Fairchild, a fellow I knew only as Lynn, and myself.



Andy Fairchild

Jacob "Jake" Lynn

They kept asking me about the cave, and I told them all I knew about it, that thousands of bats apparently made it their home, and that the entrance was quite large. That was all I could tell them because I had returned with Sam and Brown without even looking inside.

When we arrived at the spot both Andy and Lynn were quite excited. The cave was deep enough so that a rope ladder would be necessary in order to reach the bottom. I had brought back the same rope ladder we had used to extricate poor Sam, so we didn't have to waste time making another.

We soon had it securely fastened outside the entrance and then I said, "Well Andy, would you like to go first?"

"Heck, no," he replied, backing away. "You just went into one cave and said it was an interesting experience. Why are you afraid to go into this one?"

For a moment I thought I had another Sam on my hands. Then I realized he was right, and also that I had instigated this cave hunting party, so why shouldn't I lead the way?

If I backed out now, there would be no cave exploration, so, without a further word being said, I began to lower myself into the mouth of the cave.

Down, down, down I went. As I looked up I could see two heads peering down at me. They both wanted to make sure I reached the bottom safely. A

moment later my feet touched a pile of rock on the floor of the cave. It was so dark that I couldn't see anything except the dim outline of the nearest wall.

"How ya' coming?" Andy shouted down at me.

"I made it all right," I replied. "Come on down, and bring that torch that is in the back of the wagon. It's black as night down here."

Lynn ran to the wagon to get the torch as Andy prepared to come down the ladder. He came down more quickly than I did, probably because I had blazed the way and he knew it was safe.

Before I knew it, Lynn had entered the opening and was coming down the ladder. In my excitement I forgot to ask him to remain outside for safety's sake. If the rope ladder should slip or in any way become insecure, we could never have gotten out and would surely have died in the cave.

But my attention was certainly not on safety at that moment. I was much too excited about what we would find.

We lit the torch, but the cave was so large that the light of our one torch didn't help much.

Our first concern was whether or not a bear or some other wild beast might attack us. We searched the floor of the cave for tracks, but not a sign of man or beast did we see. The floor of the cave seemed devoid of any evidence of any walking creature having preceded us into this mammoth underground cavern.

Occasionally a bat would sail by, missing us by inches.

With the dim light of the torch we looked around and were struck speechless by the immensity of the great cave. We inched along over the floor of the cave which at times was rocky and difficult. We encountered large boulders and had to climb over them.

"What do you make of it?" asked Andy

"Biggest thing I ever saw underground," I answered. "Seems like we're suddenly in another world. Notice that peculiar odor?"

Lynn, who had been the most quiet of our trio, spoke up. "Animals of some kind," he said. "But I wouldn't know just what."

As we proceeded further the smell became stronger and more pronounced. Our caution increased, for I know all of us expected at any moment to see some animal lunge out at us. We kept our eyes open and, with the torch held above our heads, were able to see several feet ahead of us. Beyond that the outline of any object was too dim for us to definitely make out what it was.

The animal odor was getting stronger. We were definitely nearing something, but just what I didn't know. As a result of this uncertainty we slowed our pace, stopping every few feet to listen for any sound, yet nothing did we see or hear.

At any moment I was sure we would see two moving balls of light race toward us, which would indicate the eyes of some wild animal, but as we progressed further inside the cave our fears were unfounded.

Finally the smell became quite pronounced and at the same time we came upon huge mounds of something which was unlike the rest of the floor of the cave.

I glanced upward, and on the walls and ceiling of the cave I had the answer to the smell.

"Bats!" I exclaimed. "Millions of them. This is where they live. And those mounds beneath them are the result of their living here. Why, there's enough fertilizer there to ..."

I didn't finish the sentence. As far as the eye could see, which wasn't too far in that dim light, there were piles and piles of guano, which is the commercial name for fertilizer created by animals of this type.

Now our fears were ended, as we felt relatively safe from bats, even though there must have been millions right there over our heads. Now they were asleep, but shortly after sunset we knew they would suddenly come alive and head for the cave entrance, where they would fly off into the night in search of food.

By morning they would again form that black, funnel shaped cloud I had watched previously as they re-entered their home—a home that must have been theirs and theirs alone for countless centuries—if the huge piles of guano at our feet was any indication, and it surely must have been.

In places these piles of bat deposit reached almost to the top of the cave. Later we found that this guano reached almost a quarter of a mile in length and stretched some 75 feet in width. Some of the piles later proved to be over a hundred feet deep!

Even the crudest calculation would have shown that there was enough guano here to merit the cost and trouble of getting it out of the cave and selling it commercially. At that moment I decided to stake a mining claim on the cave.

We felt that for one day we had seen enough and were ready to head back to the entrance and call it a day. The bats were apparently the sole tenants of the cave, for we saw no evidence of any other living thing ever having invaded its dark, vast interior.

Lynn headed up the ladder first and I asked him to go to the wagon and get four small cloth sacks for me. When he dropped them through the opening, Andy and I went back to where the guano was and filled the sacks. I wanted to have the guano tested to make sure it was of good enough quality to make my contemplated mining operation worth while.

It would be silly to go to all the trouble of getting this guano to the surface and into Carlsbad, only to find that it was of inferior quality and not worth the cost and trouble of extracting it.

"How are you going to get this stuff up to the surface?" Andy asked as I was filling the sacks.

"By Ned, I don't know, but there's a way, and I'll find it."

As we wound our way back to the rope ladder, Andy and I each had two sacks of the guano, one in each hand. We made it up the ladder and out into the open again.

"Now I'm going to do something about marking this place, to show I've been here." A short distance from the entrance to the cave was a mescal pit, possibly left by Indians.

"Here, Andy, give me a hand," and with that we began gathering several large stones and placed them one on top of another until we had a pile some four or five feet high. This would have to serve as a marker until the claim could be completed.

"Well, what do you think, Lynn? Was it worth the trip?" I asked.

"Didn't know there were so many bats in the world," he replied.

"The size of that cave is unbelievable," was Andy's comment. "How big do you suppose that cave really is, anyhow?"

"That's anybody's guess. Maybe we saw it all today, and maybe we saw only a small part of it," was my humble answer.

"Bet that entrance is a busy place at sunrise and sunset," Lynn observed. It was clear to see he was more interested in the millions of bats than the size of the cave. "I'd like to get more torches and see more of that bat cave."

"Some day we will," I replied. "Right now we're heading back to Carlsbad. I'm not going to waste any time filing a mining claim."

And with that the three of us climbed aboard the wagon and headed down the hill.

4

The First Guano Is Mined

News spreads fast. Soon everyone in Carlsbad had heard about the big cave that Lynn, Andy and myself had been describing. Many of the people thought it was just a story we had cooked up and wouldn't believe us. Of course it was known there were caves in the Guadalupe Mountains in that area of the state, but no one thought the cave we described could be as large as we claimed it was.

My first job was to take care of my claim and I promptly attended to this important matter. I got the necessary papers in Carlsbad and returned as soon as I could to the big cave. I filled out one set of papers and placed them in the monument Andy and I had built. The duplicate set was taken back to Carlsbad and filed with the County Clerk.

Now everything was all set for me to begin operations. The guano, I learned, was commercially acceptable, which meant that I would have a market for the product, and that my time in bringing it out of the cave would not be wasted.

It took me several weeks to get all the details straightened out and find a crew of fellows who were interested in working on the job. It was more than simply going to the cave, taking out the guano, and hauling it back to Carlsbad.

Place Location Note

Note is hereby given, that the undersigned Abijah Long of the United States
in compliance with the United States and local Mining Laws have this
day located the following described Place Mining ground containing Twenty (20)
containing 1/2 acre. The ground is situated about 1/2 mile from the
above named up for 1/2 mile is below the office is west of the Mineral about
25 miles South west of Carlsbad N. M. and near the Great Canyon road about
10 miles South east of the Carlsbad Spring situated in East of N. M. contain-
ing valuable deposits of Iron Ore and other valuable minerals situated in East of
the Mining District County of Siskiyou State of New Mexico.
This is given in full compliance with the local Mining Laws.
Witness my hand and seal this 1st day of May 1880.
Abijah Long
N. M.

Abijah Long's original claim to the Carlsbad Caverns

[Click here for original negative version.](#)

[illegible]

Amended Notice of "Big Cave" Mining Claim (Handwritten)

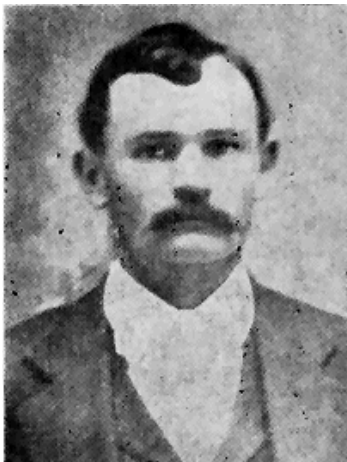
Notice is hereby given to all whom it may concern that A. Long, a citizen of the United States, over the age of twenty-one years, located what is called the "Big Cave" Placer Mining Claim which is hereinafter particularly described on the 28th day of March 1903 and thereafter to wit: on the 16th day of June A.D. 1903 filed the notice of the location for record in the office of the probate Clerk and office Recorder of Eddy County, New Mexico. Which notice was recorded in Book 1 of Records of Mining Claim at Page 149, and whereas, appears from said notice, as recorded, that the same is defective, in the fact that it fails to properly describe said Mining Claim as located. Now therefore for the purpose of amending and correcting, said location notice, I the said A. Long do hereby give notice that I located that certain Placer Mining Claim of ground in accordance with the Statutes of the United States, and the laws of the territory of New Mexico, the said 28th day of March 1903 in the County of Eddy, territory of New Mexico, which is by me called the "Big Cave" Mining Claim and which is particularly described by meets and bounds as follows, to wit: Beginning at a stone mound erected by me at a point about 20 feet west of the entrance to what is known as the Big Cave. Situated about in a Southerly course from Walnut Canyon and about one mile therefrom and about 25 miles in a southeasterly course from the town of Carlsbad, and about one mile in a southeasterly course from the Donahoe Springs, in the foothills of the Guadalupe mountains. From said stone

mound, running north 300 feet to a stone mound, which is the northwest corner of the claim, running thence east 1500 feet to a stone mound, which is the northeast corner of said claim, thence South 600 feet to a stone mound, which is the southeast corner of said claim, thence running west 1500 feet to a stone mound, which is the southwest corner of said claim, thence north 300 feet to the place of beginning. Covering in all about 20 acres of land. Said claim contains valuable deposits of Petroleum, Oil Clay, Building Stone, Guano, Phosphates, and other kindred substances situate in the Eddy County Mining District. This amended notice is made by me this 19th day of November A.D. 1903.

ABIJAH LONG

Filed for record November 19th 1903 at 3:20 o'clock P.M.

W. R. OWEN, Probate Clerk and Recorder.
By N. CUNNINGHAM, Deputy.



Charles Hannsz
Supervisor of
transporting guano to
Carlsbad

Much work had to be done before we could even touch the guano. The road up the hill, if it could be called a road at that time, had to be cleared and widened and in places leveled because it was so steep.

I had hired eight or ten boys in Carlsbad to assist in this work, and it was all carried out under the supervision of my brother-in-law, Charlie Hannsz. It proved to be no easy task to remove the large boulders, dig up thickets of cactus and other desert shrubs, and make the roadway smooth enough to allow heavily loaded wagons to pass smoothly.

At best the road was not very satisfactory, but it was the best we could do, so we had to use it.

Now that the road was done, the next part of the operation could begin. Since the entrance to the big cave was some distance from where the guano was located, I tried to figure out some way of making that haul much shorter.

On one of my trips into Carlsbad for supplies I heard of a man by the name of Victor Queen who might fit into our company. As a result I looked him up, since I had heard he had had mining experience.

"Had any experience mining guano?" I asked him.

"No," Victor replied. "But it doesn't matter much what you take out of the ground—the problems are pretty much the same."

"Well," I explained, "we have to haul this guano quite a ways to the exit, and I think there must be a way to abolish all that trouble."

"Can't you sink a shaft right over where the guano is located?" he suggested.



Victor Queen J. H.
Lockhart

"It might be a tough job to tunnel down," I mused.

"Just use a bit of dynamite," was Queen's solution.

"I've never used the stuff, and I sort of hesitate to experiment. What do you know about it?"

"Used it in Mexico quite a bit. Saves a lot of digging time, and there's no need for danger if a man knows what he's doing."

"Believe that might be our answer. How about joining us?"

Next day Victor Queen was deciding just where to dynamite a hole through to the guano. He was assisted in this powder work by Arthur Sinclair and Johnny Forehand. Among the others who were loyal workers were John Queen, Victor's brother, Wayne Crowder, and John Lockhart.

Blasting the hole proved to be no easy task, but eventually we achieved our goal. This first shaft was sunk in 1903, and it seemed then that the major part of our effort had been completed.

Getting to the guano was one thing, and getting it out was another. When we got the shaft sunk we found we would have to build a platform underneath in order to more easily raise the guano from the floor of the cave to the shaft entrance.

What appeared to be a simple job at the start proved to be rather difficult and perilous, but under the supervision of Johnny Forehand and Arthur Sinclair the platform was completed and it turned out to be a good job.

Once again it seemed that the actual mining of the guano could begin, but again we had miscalculated. Attaching a pulley and a rope to the top of the shaft, we lowered an iron bucket, shoveled it full of guano, and the first load came out.

At this rate it would take forever to get a pay load to Carlsbad. We had to find a faster method.



Wayne Crowder, Sr. John Queen J. F.
Forehand

By this time my finances were dwindling fast, and in order to continue I found I was compelled to seek outside assistance. Many people hesitate to have anything to do with a mining operation, figuring there is too much risk to merit any financial backing. But with us the mine wasn't guesswork because we could see the guano and knew there were many tons of it waiting to be brought to the surface and shipped to San Francisco.

As a result, I soon made a contract with the Ramsy Brady Company of Carlsbad whereby they would assist financially in the backing of the undertaking in return for a half interest in the profits.

With that important detail out of the way I was ready once more to give my full attention to extracting the guano which had so far remained in the big cave in spite of all our endeavor.

Now we were ready to complete the final stage of our operation.

Since the iron bucket was much too slow, we decided some quicker means must be found. We held a consultation and several ideas were suggested. The best seemed to be to build a track out of 2×4 lumber and construct a small car or wagon to run on this track. Matt Ohnemus of Carlsbad was assigned to this job, and he built us a strong and sturdy car which I was sure would serve us well for a long time.

It was similar to the hand cars used on the railroad, except that we were forced to use wooden wheels, the iron variety not being obtainable.

Our system was now changed. We would scoop up the guano in shovels and place it in sacks. Eight or ten sacks could be loaded onto the car and hoisted to the surface.

The sacks, of course, were sewed across the top before being placed on the car. Johnny Forehand proved to be a master at sewing these sacks. He was fast and accurate, and many of the other fellows learned to speed up their sewing by watching Johnny.

I was really quite fortunate in having acquired such a fine bunch of fellows to work on the operation. They all seemed to adapt themselves quickly and well to any job assigned to them, and furthermore they seemed to take an interest in their work.

One of the big reasons for this was that all the men knew there wasn't too much work in that section of the state, and therefore they were eager to keep the jobs they had. Also, the cave seemed to possess a certain degree of mystery for the men and they felt a certain amount of daring and adventure connected with the whole operation.

Working with guano can't be classified as a glamorous operation. The very nature of the substance—bat deposit—has a rather disagreeable odor and it takes a while to get used to it. Also, there wasn't much circulation of air in the big cave with the result that the smell had been bottled up there for centuries and we were getting the full effect of it.

From time to time fellows would quit or leave for one reason or another, and whenever a new man started in he would comment on the odor, and then we were conscious of it, but otherwise we grew so accustomed to it that we didn't notice it.

The men were human, and just like men everywhere, I guess. They worked hard during the day, but during their off hours they liked to partake of the pleasures men normally like.

Of course we were 28 miles from Carlsbad, so the fellows couldn't go that far to spend their evenings. They were, therefore, forced to provide their own entertainment in our own camp.

Life in and around the big cave was never boring. There was something happening every minute. New men, it seemed, were the target for pranks, and my boys soon became adept at staging them!

5

Life at the Camp

New men were always nervous about going into the cave. A mine is one thing, but a cave! And people back in Carlsbad were continually enlarging their yarns about the adventures we were having in the "big cave" some 28 miles away. As a result, all new men who came to work had illusions about this mysterious cavity in the earth where bats dwelled by the millions.

The boys thought new men were fair game—for anything. Take for instance the time a new man came out to work at the cave and the boys decided to have some fun.

They asked the tenderfoot if he would go down into the cave for a shovel, and the man obeyed. As soon as he reached the bottom, a voice boomed out of the darkness: "What are you doing down here?"

The man, not knowing anyone else was underground at the time was so scared he could hardly talk. Finally he uttered, "Why, I have—a—well—I have come to get a—a shovel!"

"Now lookee here," boomed the voice again. "I've lived in this cave nigh on to 50 years, and now you guys think you're going to take it away from me. I'm going to kill you!" With that a gun fired at close range.

The new man, scared out of his wits, fairly flew up the ladder and headed for open country, as fast as his legs would carry him. As far as I know he is still running, for none of us ever saw him again.

Another time we had a young Mexican boy working in the cave sewing sacks. One day he got upset over something and suddenly decided to quit, saying, "Me voy, me voy!" (I am going, I am going.)

It was really nothing serious, and we pleaded with him, trying to explain to him that the matter was not important, that we liked him and wanted him to stay as we needed him. Furthermore, it wasn't easy to get men to remain at the cave.

Our pleadings were of no avail, for he started out and we couldn't stop him. He jumped on the car and began to pull the ropes and in that way propel himself to the top of the cave.

Ropes in those days were not too well made and they wore out fast. He didn't know, and, in fact, we didn't either, that the rope on the hoist was ready to snap.

Up he went, almost to the top, and then—snap. Down came car, Mexican boy and all, right smack into a waiting load of guano. He was covered from head to foot—just a leg sticking out.

He didn't move, and we were sure he had been killed. We started digging and soon uncovered a very much alive but very scared little fellow. The guano had cushioned the fall.

Apparently he thought we somehow caused the fall to prevent his escape, for he never tried to run away again, and soon became one of our best workers.

Incidents like this were not good for the morale of the men. The pranks, of course, were harmless as far as danger was concerned, but natural risks were always present.

Sometimes the pranks were meant to be harmless, and they were to the men, but it worked a hardship on us who managed the mine. I remember one day Victor Queen, who at that time was in charge of operations, had to go to Carlsbad for supplies.

"Johnny," he said to his brother, "I'm leaving you and Wayne Crowder in charge while I'm gone. Keep the men busy. I'll be back as soon as I can make it."

Wayne was a good worker and always reliable, but like lots of men he enjoyed a good laugh and often went to long ends to get it.

"Johnny," he said, "you engage the Mexicans in conversation. I'm going to fix up a ghost and we'll have a little fun with them."

While Johnny Queen was talking to the Mexican workers about everything he could think of, Wayne made a large paper man, then tied a string to it and dangled it from the roof of the cave in a dimly lit section which hadn't been explored too much.

The Mexicans were quite superstitious about the cave, and Wayne knew he would get quite a rise out of them when they saw this ghost-like figure rising out of the semi-darkness.

He didn't have long to wait. Signaling to Johnny that the scene had been set, the conversation stopped and the Mexicans were sent into the cave in the vicinity of where the ghost had been rigged.

Panic broke loose. Every Mexican in the group quit, scampering out of the cave and heading for Carlsbad. They were scared. But so were Johnny and Wayne. They knew how hard it was to get men to work in the cave and, with this mass exodus of workers, they would be in for a hard lecture from Victor when he returned and found what had happened.

They pleaded with the frightened men, but it was useless. They had seen that white figure with their own eyes. They had heard the cave was haunted; now they knew that what they had heard was true. Come back to work? Not in that cave! Some picked up their belongings, while the others were in too much of a hurry to get out of that section of the country.

When Victor returned and learned what had happened, he nearly fired John and Wayne on the spot. Needless to say, John and Wayne had learned their lesson, and they were much more careful about what they tried on the workers in the future.

On one occasion a man by the name of Priest came out from Carlsbad to see the cave and learn how guano was being taken out. The boys were always ready to oblige anyone like this, so they told him to get on the car and they would take him down.

The car hadn't gone very far when the steel cable on the drum slipped a bit and the car suddenly dropped about a foot.

Priest was so frightened he began to pray. The boys thought that was quite an incident, seeing a Priest pray on a cable car going after guano, and they always got a bang out of telling it.

That car contributed its share of incidents at the cave. One other time we had just received several hundred empty sacks and were preparing to send them below where the men could fill them with guano.

Generally the workers in the cave would walk the ladders when they wanted to go into the cave or come out. But this time there were three boys who decided they would get a free ride as long as the car was going anyway with the sacks. So they jumped on.

Sitting gaily on the sacks they appeared ready for a thrill, so they called out to the whim man, "Let 'er go!"

He did.

The boys were asking for a fast ride, so he decided to give it to them. He forgot to take into consideration that the weight of the several hundred sacks plus the weight of the three boys was too much for the car, and as it gained momentum in its downward plunge the brake gave way.

The car was now falling freely of its own weight, and in an instant would crash at the bottom, totally wrecking the car and possibly killing the three boys.

I happened to see the whole thing, being right there at the time, and before I could think what was happening I had instinctively grabbed a 4×4 timber close by and rammed it into the whim.

Smoke billowed from the 4×4 as it hit the whim. The car slowed up just as it hit the bottom.

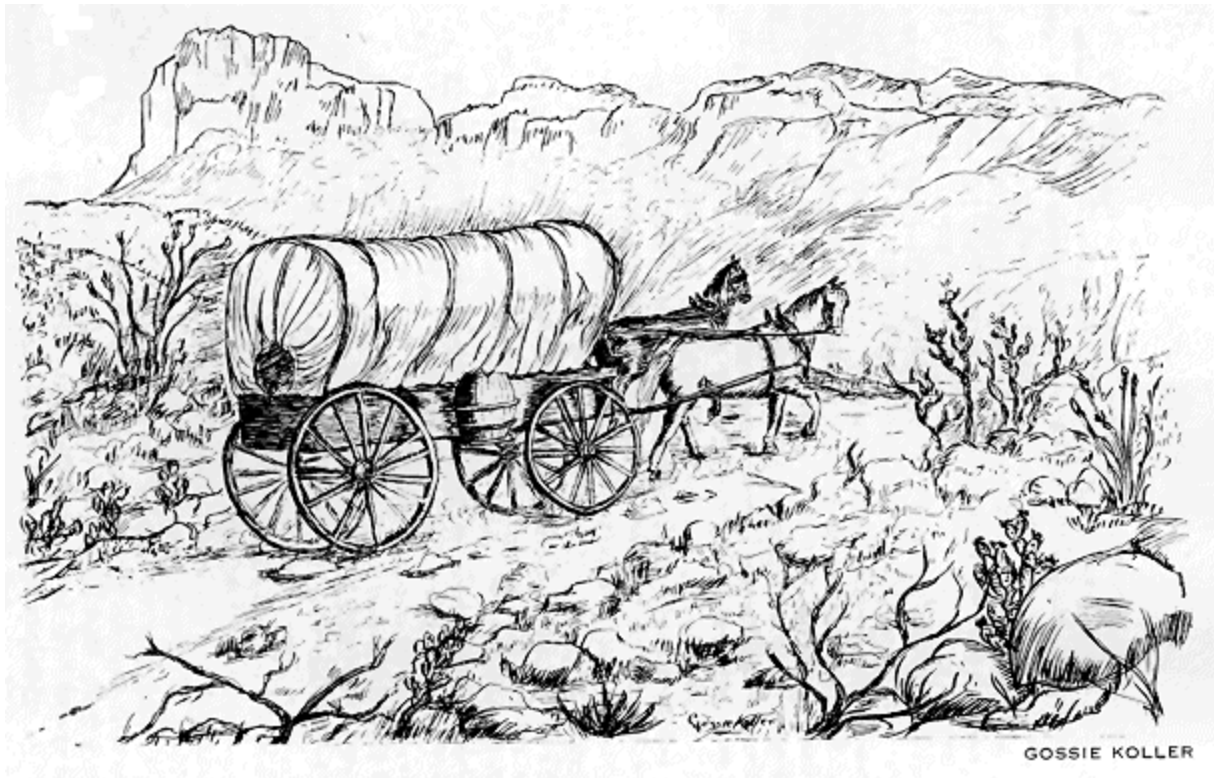
From our position at the top of the shaft we couldn't tell how hard the car had landed, nor could we tell if anyone had been badly hurt.

Just as I was about to call down, I heard one of the boys call up, "Can't you take it a bit slower? We'd at least like to have a little conversation on our way down."

I knew from that jesting that there couldn't be too much damage, otherwise he couldn't joke about it. We soon learned that no damage had been done since I had injected the 4×4 into the whim just in time. There were a few bruises, but that was all.

Life at the cave was somewhat rugged at first because we didn't have many of the comforts of home. At first we would sleep in the wagons, then we brought out some canvas tents from Carlsbad and this made living much more comfortable.

As operations continued I had hoped to build some sort of small house and move my family out to be there with me. But my time was so busy in the early phases of the operation that I had little time to think of anything but getting the guano out of the ground.



Artist's conception of Long's move to Big Cave area.

After a while things became more or less routine and I could then think of some more permanent living quarters for myself and family. My brother-in-law, Charlie Hannsz, also had more spare time now that the roadwork was done and the initial construction chores had been completed.

Since he was a fine carpenter, I let him have the job of building more permanent living quarters.

We found the most satisfactory type of housing was what we called tent houses. These consisted of boxing plank for the floors, with the wood also extending about half way up the side or wall of the house. The upper half of the wall was canvas, as was the roof.

These houses were relatively easy to build, and served our purpose adequately. After the first house was completed, we built another for my father, A. B. Long, and my mother. Then we built another for Charlie Hannsz and his family, making a group of three houses.

My wife made our place quite comfortable inside. She was an immaculate housekeeper and she had the knack of knowing how to make our little tent house look like a home. I felt quite contented living there.

The food problem was a difficult one, for all our supplies had to be brought out from Carlsbad. At first my wife did all the cooking for the men, but this job soon became too much for her, especially as the operations at the cave grew and more men joined the force.

Eventually the men had to make other arrangements for their meals. Most of them did their own cooking, or they would divide up into small groups and take turns cooking for the rest. The Mexicans would cook their own meals off by themselves as they seemed to prefer a different menu from the rest of us.

This section of the state was good for cattle grazing and this was quite an industry in itself at the time. There was a man by the name of Simms who had quite a herd near the cave and sometimes during the night we could hear these animals near our camp.

One morning, as the boys were entering the cave to go to work, they discovered that one of the cows had fallen into the cave. In trying to reconstruct what had happened, it looked as though one cow had hooked another, causing one to fall into the cave.



Artist's sketch shows family cabins set up near cave's entrance.

The boys butchered the animal right there, carrying the cow out piece by piece. I have often wondered if Simms ever knew that we were enjoying fresh steaks from one of his herd!

Often on evenings or Sundays the boys would get together and hold what they called a Kangaroo Court, thus helping to pass the time away. First, they would select one of their number to serve as judge.

Then, from the group, one of the fellows would be selected for some misdeed he had committed while on the job that day. After the testimony was heard, his fellow workers, who served as an impromptu jury, would decide the verdict, which was most always "guilty," and the judge would impose the sentence, such as requiring the guilty culprit to take a small pail and a coal oil lantern and go back into the deep recesses of the cave alone and return with the pail full of water.

Entertainment during the evenings varied, depending upon the mood of the men and what their particular likes and dislikes were. Card games were

always popular, and a hot game of poker was the favorite with most of the men.

Many times after they had finished a busy day in the Big Cave they would roll out a tarpaulin, deal out the cards, and proceed to have a gay time.

More than once these games would last all night, or until one man would win all the money and the rest had no more cash to play with.

Johnny Queen must have been lucky at love because he certainly wasn't lucky at cards. In practically every poker game he would soon lose all his money and would be out. On cold nights the fellows found the playing not very comfortable, and after a while they got the bright idea of asking Johnny to build a fire for them, since he was only watching anyway.

This Johnny did, using sotel^[C] for fuel. Sotel is a type of desert plant that grew in the area around the cave. With a warm fire to comfort them, the games would go longer than ever.

[C] Sotel—a desert plant (dasytirion) having a stiff stalk and crowded leaves.

Johnny, however, couldn't understand why he should gather fuel and tend fire while they enjoyed all the fun, so he got the bright idea of charging the players a dollar. They were glad to pay. Needless to say, there were many nights when Johnny ended up with more money than most of them did!

6

The Big Cave Is Explored

After the mining operation had been under way for some time I found I wasn't so busy and had a little free time to myself to sort of rest up and think of something besides getting the guano to the surface and off to the Hawaiian Fertilizer Company of San Francisco.

More and more I began to wonder just what this Big Cave was like beyond our immediate area. Many of the men also wondered, and often suggested that some day we form an exploring party and see what it was like.

We decided to do just that. A day was decided upon and the necessary preparations and precautions were made. Of course, we had no idea of what lay ahead of us, nor how far our journey would take us, and I reasoned therefore that it would be better to carry too much with us than too little.

First of all I rounded up all the available lanterns that weren't otherwise in use. Then I got all of the candles we had, a supply of matches, and all of the cord in the cave. We had a lot of cord because it was the cord we used to sew the guano sacks.

We also carried a supply of food, for we meant to really go a great distance. Of course no one had to go, but most of the men wanted to see what mysterious wonders lay beyond the site of our immediate guano operation.

Fortunately there were a couple of men who didn't seem interested, preferring to remain outside. This was all right with me, for if any unforeseen emergency should arise, we could depend upon them to send for help.

We decided how long we would be gone, and I told them that if we were not back by a certain time, they were to organize a search party and come looking for us.

I told them that we would light the candles and place them at intervals along the way, and also we would string the cord along the entire distance, both methods being used in order that we could easily and quickly find our

way back, as well as to assist a rescue party in following our trail should such prove necessary.

With all of these precautions being made, the suspense of our undertaking grew, and the men were quite excited and eager to get started, yet I was in no hurry to proceed until I was sure everything was in readiness, for although we were to be gone only a matter of hours, no one knew what lay ahead and I preferred to be well prepared for any emergency.

So, at our appointed hour our journey of exploration began—the first journey into the Big Cave—the first time Man had ever set eyes on its matchless wonders. How little we knew then that in the years that would follow millions of others would enter this cave to view the beauty and grandeur that Nature had been in the process of creating for countless centuries.

At times the going was pretty slow due to the uneven floor of the cave. Big boulders often had to be moved to one side, and when they were too heavy or were too well lodged in the floor to permit moving, we had to climb over them.

Generally the direction of travel was downward as the Big Cave seemed to go deeper into the earth. We were going in a westerly direction and the cave seemed to take on a maze of large connected rooms.

We would enter one and then, as we would pass out the other side, seem to enter another.

Many times we stopped in bewilderment of the beauty that lay before us. The fellows in the party often kidded about the trip and were pretty much in gay spirits, but as they would enter these magnificent rooms which man had never seen before, they became very quiet, letting their eyes soak in every detail that the light of our lanterns brought out.

We all seemed humble amid God's handiwork, and I noticed many of the roughest men in the party became suddenly reverent as though they might be entering some great cathedral.

I never knew exactly how the other men felt about all this strange exhibition of beauty, but I could see it had some effect upon their lives. A rough bunch of men usually don't say much about this sort of thing.

As for me, I wondered if anyone on the outside would believe our stories when we told them what we were seeing. Actually, I felt it was a big dream, and that I would soon wake up. I had heard of caves before, but I had never heard of anything this large or filled with as much beauty.

All of us felt that each turn would be the end and we would be able to terminate our trip and return. But each time we would discover a deep shadow which would turn out to be another tunnel or entrance to another large cavity or room, often more colorful and appealing than the one we were leaving.

At times there would appear to be no further trail. The boys would flash their lanterns around and find an outlet, sometimes so high above us that we couldn't reach it without a ladder. Then we would find another on our own level and we would be on our way again, leaving a world of beauty for another fully its equal.

In one place we came upon what looked like an iceberg, but of course it was a rock formation. In another we came upon a green pond of water.

Everywhere we saw stalactites formed through countless years of the dripping of water, each drop leaving behind its minute portion of calcium or other chemical composition which eventually forms the pendant. The fellows constantly referred to them as icicles, since they resembled them so exactly.

Underneath many of the biggest ones were the stalagmites, the "rising statues" formed by the drips from their counterpart above. Every one, it seemed, had a style and shape all its own.

The colors were in a class by themselves. We saw all the colors of the rainbow, from pale pinks to deep maroon—from pale sky blue to dark purple. Many had a softness which no artist could duplicate. Others were as bright as a Mexican cafe, and the Mexican members of our party took an extreme delight in seeing these vivid hues. Apparently they had missed, in America, the bright colors which are so abundant in their homeland.

Almost any formation known to man could be seen as we inched our way along. Castles, monuments, animals of all sorts, a gremlin, the devil, a bridal veil, totem poles, all of these and hundreds more were easy to

visualize everywhere around us. And with but a little Imagination we could see many more.

It was time to turn back, and I could feel the reluctance of the men in the party to do so, but if we did not arrive back at our appointed time a search party would start after us, and I saw no reason to create false fears with those on the outside.

The return trip was much faster than our trip in, simply because we knew what to expect. Also, we did not stop so long to look at the magnificent splendor, and also because the candles and string told us exactly where the trail led.

It was interesting to note, however, that as we would enter a room from the opposite direction from which we had originally entered it, it appeared much different, sometimes so much so that we thought we had not entered it previously. Of course, the candles and string proved we were wrong, but the difference in perspective was amazing.

When we finally arrived back at the entrance the members of the party spent hours telling the others what we had seen. It was interesting to hear the different descriptions from the men and to see how each man had noted something different yet of course we had all been along the same route.

After this we often took trips into the cave and would explore different rooms and tunnels which we hadn't seen before. It seemed that there was no end to this maze of underground caverns.

Accidents were relatively few on all of these trips, and I think it was because the fellows all realized that if any of them got hurt it could be serious. The trip back would be a hardship on those who might be required to carry an injured man back to the entrance.

When my young son, Jodie, (Joe N. Long), was about 5 years old, he began to take quite an interest in our conversations in which we would describe the many incidents that took place daily in the Big Cave.



Abijah Long



Joe N. (Jodie) and Andrew B.
Long Long

(Taken about 1904)

Finally Jodie asked my father to take him into this big hole in the ground. We didn't think it safe to take such a small child into the cave, but Jodie kept insisting.

However, Jodie won out. One day his grandfather led him to the cave entrance, the one where visitors now enter, and took him inside. From that day on we were besieged with all kinds of childish questions.

The incident is worth mentioning because, as far as I know, Jodie was the first child ever to enter the Big Cave.

We never did see any forms of life in the cave, other than the millions of bats which made it their home, but on one of our trips into the inner chambers the boys found a large bone. They brought it out to examine it more carefully. It was much too large to be a human bone, and when the boys tried to break it they found it was very hard, resisting for a while the blows of their sledge hammer.

The bone apparently was in a state of semi-petrification. We all speculated as to how it could have gotten into the cave, and every man had his own theory. Some said an animal must have fallen into the cave, such as Mr. Simm's cow had done, but lived to stumble on into the cave where he starved to death.

The fellows thought this theory unlikely because if it were true, other bones would have been found at the same spot.

Another theory was that a cave man centuries ago had killed some wild beast and had taken a leg of meat into the cave where he made his home. Meat and man had long since ceased to exist, but the bone remained.

At least we couldn't offer any sound reason against this theory, but it seemed almost fantastic to accept it.

The only other theory which seemed at all plausible was that some wild animal, such as a bobcat or lynx, had at some time lived in the cave and had brought in the leg of a deer or moose and left it there after having eaten the meat from it. However, there were no tracks of any kind in the Big Cave, hence that theory had little support.

How the bone got there still remains a mystery.

7

The End of the Beginning

After the guano operation had been going on for some time I could see that it wasn't turning out to be as profitable as I had originally thought it would be. With the vast quantity of guano under ground, and with our efficient method of extracting it, I wasn't able to realize what I felt was a sizeable profit from the time and effort involved.

Of course there were lots of factors that entered into the operation which do not meet the eye. For instance, when the guano was brought to the surface it was filled with moisture, and I was paying the freight charges to San Francisco by the ton.

Yet when the guano arrived in San Francisco, the Hawaiian Fertilizer Company would pay only for guano that was dry. So, when I found I was paying freight on unwanted moisture, I stopped that in a hurry.

I built a series of drying racks or platforms in the western section of Carlsbad and let the guano remain there until it had dried out. This kept a pay load from reaching market and was one more operation which ate into the profits.

The freight to San Francisco at that time cost me \$9 per ton. The guano was \$2.25 per unit, \$1 for phosphoric acid, and \$1 for potash per ton.

One answer would have been to find a closer market, as that would have cut down the high cost of freighting, but there was no other market, so we were forced to continue sending the guano to San Francisco.

With some products the manufacturer sets his price, based on cost of production, but with us we had to accept the market price at the time, or else not operate.

When these problems were becoming more complex and I couldn't see any good solution, a man by the name of Patterson who lived in Carlsbad offered me \$500 cash for my equipment and the work I had done at the Big Cave. I immediately accepted his offer.

And then, what did he do but turn right around and hire me as foreman! So, for some time after that I continued the operations at the cave, but of course not as owner.

It seemed as though the extraction of the guano was not intended to be a paying proposition, for the Big Cave changed hands many times after that.

Little did any of us know then that the Big Cave, known in later years as the Bat Cave, would some day be worth millions as a sight-seeing attraction and would completely dwarf the income derived from the guano operations.

Of course that eventually came about, beginning in 1923. President Coolidge proclaimed the area Carlsbad Cave National Monument in October of that year, and a great amount of publicity during the years that followed prompted thousands of tourists from all over the world to come to see what some writers have termed "The Eighth Wonder of the World."

Former Owners of Carlsbad Caverns

1905—Santa Fe Railroad. (Owned east portion over Bat Cave. 40 acres.)

1906—C. F. Hagan sells one half to H. F. Patterson of Carlsbad.

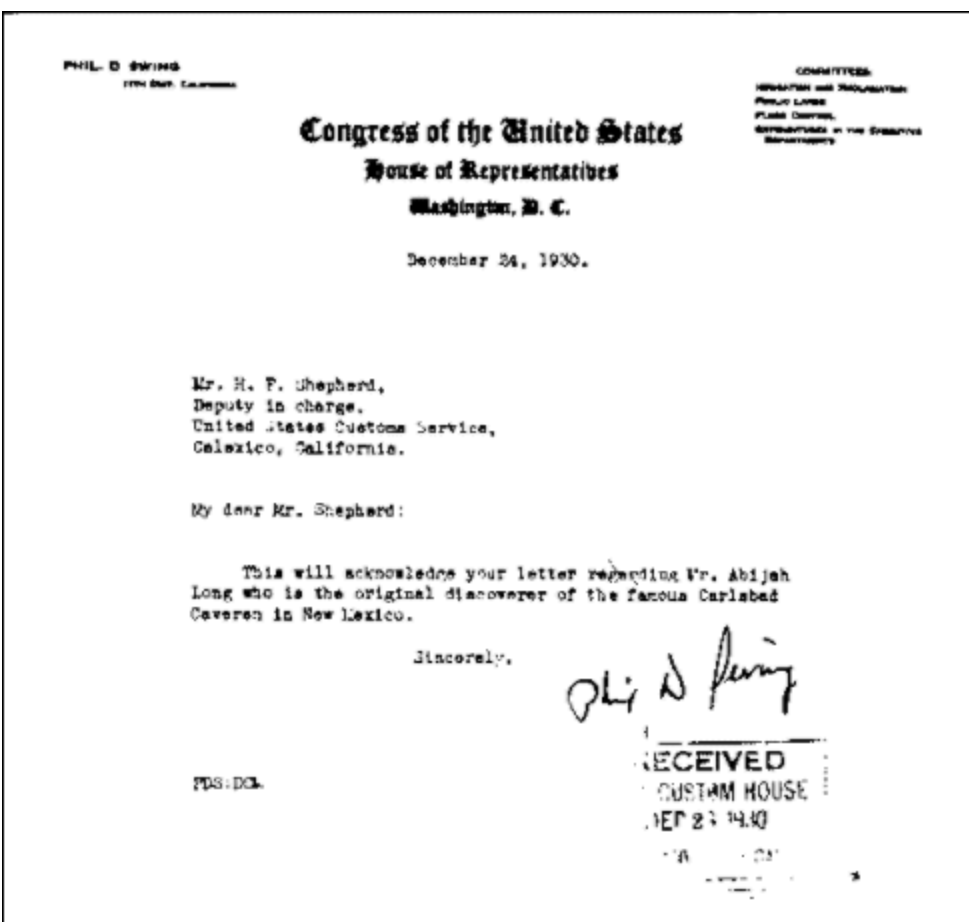
March 26, 1906—T. W. Teague of El Paso.

April 19, 1906—The El Paso Guano Fertilizer Company.

April 21, 1908—G. M. Cooke.

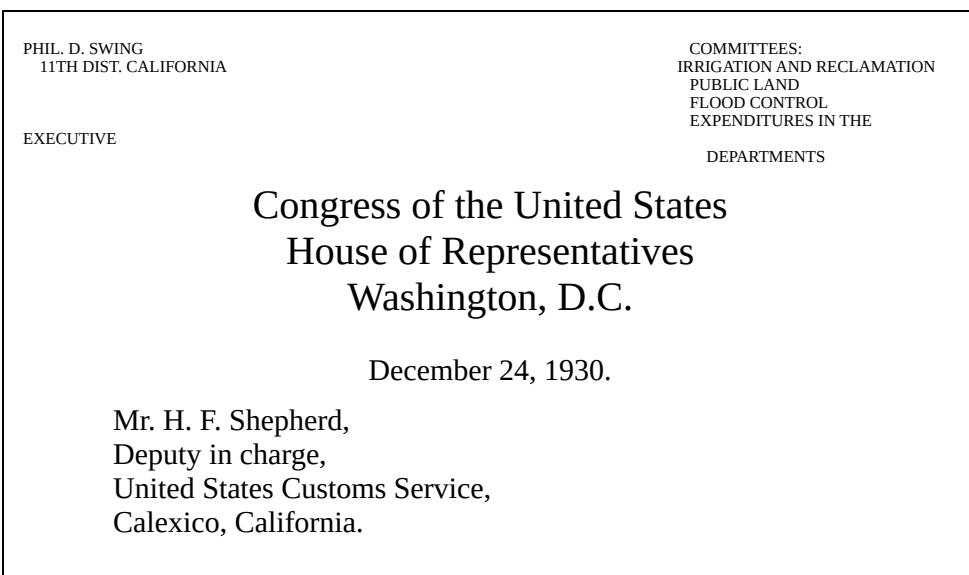
September 12, 1909—J. D. Lanford sells to Carlsbad Guano Fertilizer Co.

T. A. Blakely held a patent to the east wing of the Carlsbad Cave, where the guano beds were worked extensively. In 1942 the Government purchased this tract from Mr. Blakely.



Letter from U. S. Congress recognizes Long as cave's discoverer

[Click here to see original negative image.](#)



My dear Mr. Shepherd:

This will acknowledge your letter regarding Mr. Abijah Long who is the original discoverer of the famous Carlsbad Cavern in New Mexico.

Sincerely,

[Signed: Phil. D. Swing]

PDS:DCM

Well, that's my story! That's how it all began. I'm glad to see that the Federal Government has taken over supervision of the Big Cave and has made it available to people from every land in order that they might enjoy its unsurpassed beauty which first greeted us 'guano miners' back in 1903. It is gratifying to me to have been a small part of the early history of the Big Cave, now more appropriately called "Carlsbad Caverns" and known all over the world.

I am also grateful for the privilege of having known and worked with these men, many of whom grew with Carlsbad and turned out to be prominent citizens of the community.

Times may change, but the great caverns will remain for countless centuries to come, just as they have been for years without number. Now, in the hands of the government, their preservation is assured, so that your children and my children and their children after them will be able to witness, unmarred and unchanged, these wonders of God's handiwork.

And you'll be thrilled just as we were, when you see them for the first time, and you'll marvel, as we did, that anything in the world could exist, so strange, so large, so utterly beyond description.

Man may build his temples and his shrines, but as you stand there, a mere midget in any of the several mammoth rooms, you will feel a certain humbleness come over you, and you cannot help but feel closer to your God.

Part II
THE
FIRST 60,000,000 YEARS

By JOE N. LONG

1

The Beginning

How old are the Caverns? When did they begin to form? Are they growing larger today? What changes, if any, are now going on?

These are questions which everyone who becomes interested in the great Carlsbad Caverns wants to have answered sooner or later.

To some of these and many similar questions the answers are difficult to obtain, for when the Caverns were being created no man was around to watch the process and to report his observations first hand. As a result, we have only the word of geologists and other scientists who have specialized in the study of the earth, its formation, and the changes that have come about through the countless ages since our world began.

No one knows how old the Caverns are. All man can do is to estimate, and he bases these estimates on a study of conditions he finds within the cave itself, in the surrounding countryside, and from a general knowledge of the earth and how it has evolved.

Probably the area itself began to form about 200 million years ago, during the Permian period of geologic time.

The area at that time is supposed to have been either an inland sea or a shallow extension of the ocean. During this Permian period the earth's surface was changing. Mountains were rising and the waters were receding, thus greatly enlarging the land areas. Amphibian life was on the wane, and reptiles began to appear.

Great thicknesses of limestone deposits were made during these countless centuries, some in the form of a reef now known as the Capitan limestone. Contemporaneous rock behind the reef is called the Tansill formation. It is in these rocks that the Carlsbad Caverns are located.

Since Permian time geologists think the seas may have risen again and covered what is now New Mexico, leaving sediments that have been largely removed. About 60,000,000 years ago, during the Cretaceous period when

coal was being formed elsewhere in the world and dinosaurs roamed the earth, the land was uplifted, perhaps producing cracks and crevices in the limestone.

Geologists at one time believed the Caverns were caused by the action of water as it flowed down through the limestone, dissolving as it did so minute particles of the stone. Today, however, that theory has been discarded, since a more careful and detailed study indicates the Caverns to have been formed by phreatic solution, their development resulting from a two-cycle method of creation.

The Caverns can be said to be a natural cavity in the earth formed by the solution of rock by subsurface waters. Actually, there are two great geological processes involved in the formation of the Caverns. The first came about as the water hollowed out the underground chambers, and the second took place when the formations of stone were created in these underground openings.

To be explicit, we can only say that the Caverns are large crevices or cracks in the limestone which have been enormously enlarged by the constant solution of the rock into the underground water which filled these indentations.

In order to understand just what the two-cycle method means and how it can exist, we must first understand the two conditions under which these operations of nature can take place.

Water, as it seeps downward into the earth, tends to seek what we shall call its own level. This would be a point where everything below is saturated with water, the water table of the region. Above the table water works downward due to its gravitational pull. Below the table there is only rather slow movement of water.

The area above this point is known as the vadose zone, and below this point the area is the ground-water or phreatic zone.

Any farmer who has ever sunk a well knows how important it is that his well reach below the water table if he is to be assured a continuous supply of water. Well owners also know that the water table can rise or fall due to any of several natural causes, such as an abundance of rainfall for a few seasons, or any great lack of rainfall for an extended length of time. When

the water table goes below the lowest reaches of the well, no more water can be pumped, and it is necessary to extend the well further into the earth until it again goes below the water table and reaches into the saturated area where water is abundant.

The water table is not constant around the world, even though water seeks its own level. The water table might be high in one section of the country, low in another. It might be kept high by an abundance of rain, or remain low due to outlets such as springs or underground seepages.

The vadose-water area also varies, and the effects created by the rise and fall of the water table in one section of the country, for example, might be vastly different than the effects in an adjacent territory. This explains why any action of underground water in one area is not necessarily duplicated in an apparently identical area close by.

With an understanding of the two water zones, and the demarcation line between, we can more readily comprehend just how the Caverns were formed.

Geologists are now pretty well in agreement that the enclosing rocks of the Caverns were located in the ground-water zone under saturated conditions. As we already know, large cracks existed in the otherwise solid limestone.

Then, for many, many years the ground water dissolved the limestone, enlarging the cracks bit by bit, until the huge Caverns were formed. At this point the water table was lowered, and, in time, the Caverns ceased to exist in the ground-water zone. Following the lowering of the water table, the Caverns were nothing more than empty holes in the earth. Now the second phase of their "growth" was to begin.

When the ground water could no longer continue its constant gnawing action on the limestone, the Caverns had reached their maximum size. From now on they would begin to fill up again.

The second of the two cycles consisted of mineral deposit brought about by water seeping downward from the surface, carrying lime for the secondary formations which are today one of the great features of the Caverns.

Water in the vadose zone, seeking the ground-water level, would seep into the top of the many rooms and galleries. Each drop contained a minute portion of limestone.

Some drops would remain on the ceiling, where the moisture would evaporate, leaving a tiny ring of limestone deposit behind. As these limestone deposits accumulated they formed a pendant mass resembling an icicle which is called a stalactite. Drops falling to the floor would produce deposits known as stalagmites.

Sometimes the stalactites and stalagmites would join and a solid pillar would be formed.

In the rocks enclosing the Caverns are various deposits of other minerals such as iron oxide which are often carried in minute quantities within the seeping water and deposited on the stalagmites and stalactites. These minerals are responsible for the many different colors that may be seen in many places within the rooms of the Caverns.

Some of the formations have a very live and transparent appearance, looking somewhat as though they might have just been given a thorough coat of wax. Actually, these formations are still active. That is, they are still covered with water and the age-old process of depositing the small particles of carbonate of lime contained in the water is still going on. In cave parlance they are "alive", that is, they are still growing.

In the Carlsbad Caverns today about 10 per cent of the formations are "alive". Water is seeping in around them and working its wonders. It is this presence of water that gives them the polished, radiant look.

When the water no longer reaches the formations, they "die", i.e., they no longer continue to grow. They lose their sheen and lustre and take on a powdery appearance.

Geologists differentiate the two primary effects of seeping water in the creation of the formations in caves. Those that are created by water dripping from above, as in the case with stalactites and stalagmites, are referred to as dripstone formations. Those created by the flowing of water over a surface are called flowstone formations. In some rooms of the cave, formations made of flowstone are quite abundant. This is to be found mostly on the floors of the rooms, or on walls where a large volume of water is present. These formations have been referred to as masses which resemble ice that forms on a cold winter's day near a stream of water. Others have described them as "cascades frozen in stone."

An additional phase of the cave's development has been described by geologists as the period of collapse. During the centuries when the entire area was saturated with water, limestone blocks on walls and ceilings were weakened by solution, later to crash to the floor, leaving the room larger than ever. This collapsing continued after the cave became "dry" but ended once stability was achieved. Park naturalists tell us no rock has fallen within the cave for thousands of years.

So ends the first stage of the history of the great Carlsbad Caverns, their gradual growth and formation over periods of millions of years. Man was still centuries away when their beauties were being created. But were other forms of life inhabiting them?

2

Life Enters the Caverns

At the time the limestone beds were beginning to crack, which we noted was some 60,000,000 million years ago, the huge reptiles and other prehistoric animals were playing hide and seek on the surface above the site of the developing Caverns. At that time the cave hadn't yet been formed—the underground seepage of water was just beginning to find the crevices below the gypsum and rock salt and begin its erosion which was to continue slowly for so many centuries.

Recent forms of life seem to have a more or less direct bearing on the bat, and scientists remain quiet on what forms of life, if any, used the cave for their domicile between the earliest times and the arrival of the bat.

At least, the nocturnal creature is a sure resident, for he still makes his home there, and in numbers running into the millions.

But when did the bat first inhabit the cave? That is difficult to answer, except to say that it was several thousands of years ago.

How can scientists be so sure the years are so many? There are several ways in which they have been able to establish that the centuries have been numerous since the bats first discovered this haven.

One of the best is by the guano deposits found on the floor of the cave. Observations have shown that the excrement from the bats accumulates at the rate of approximately one-half inch per year. Thus 25 years is required for a foot of the substance to materialize. This would indicate it would require about 1,000 years for 40 feet to accumulate.

Unfortunately, the exact thickness or depth of the guano deposits was not carefully measured when my father first discovered them, but I have heard him describe them many times as being at least 100 feet deep in places. That would be almost conclusive proof that the bats had been living in the cave for at least 2,000 years!

It should also be remembered that as guano ages and dehydrates, it sort of packs down, so that although a half inch of guano might be deposited in a year's time, several years later this guano will have decayed and packed down to half that thickness.

This would extend the time to far more than the estimated 2,000 years, but just how much is extremely difficult to estimate.

Also, the amount of guano growth varies with the years, for when insects are plentiful, bats gorge themselves and the guano deposits are much greater than in lean years when their food supply is low, sometimes to the extent that the bats will go elsewhere for one or two seasons, at which times there is no guano accumulation at all.

At one time, during an exploration of the cave some years ago, a scientist discovered the remains of a bat sticking out of the side of a stalagmite where it had undoubtedly fallen from its perch above, having perchance died of old age.

With no disturbance to blow its remains away, it rotted there and the stalagmite grew over it, so to speak. Scientists, estimating the rate of growth of stalagmites, were able to calculate approximately how long ago the bat had fallen. The head and wing bones were barely discernable.

It is also known that the bats have moved around in the cave, for their tiny skeletons have been found in remote sections at points where the bats are no longer found clinging to the ceiling for their daytime sleep. Such findings indicate one of two possibilities; one, that some bats might have gotten lost in the cave and died there, away from their normal resting place, and, second, that previous openings have since been closed, due to various earth movements, forcing the bats to use new openings and thus change their location within the cave.

Scientists place little credence on the former possibility because bats seldom get lost. They can find their way in any dark cave or passageway and know how to get out the same way they went in.

The second possibility is the most likely, that centuries ago there were other openings to the cave through which the bats entered and spent the daylight hours. As old age overtook them they would die and fall to the floor. Then, as subsequent earth movements closed the openings, the bats would find

new doorways to their daytime abode. This also accounts for large quantities of guano which were found in portions of the cave where no bats were to be seen sleeping on the ceiling above.

There are eighteen species of bats to be found in New Mexico, and thirteen are known in the Carlsbad region. In the United States as a whole, scientists have classified over 251 varieties of these winged mammals.

Of the thirteen found near the Caverns, eight are known to have made use of the cave at one time or another. Old skeletons have been found in remote sections of the cave which have been identified as those of the Large Pale Bat (*Antrozous pallidus*), wing bones of which were found in one of the innermost rooms; the Big Brown Bat (*Eptesicus fuscus*), a skull and wing bones having been discovered in an inner room; House Bat (*Myotis incautus*), several skull fragments of which were found in one of the lower rooms of the cave; Red Bat (*Nycteris borealis*), two old skulls of which were found on the floor of one of the deepest rooms; Cave Bat (*Myotis velifer*), evidenced by imperfect skull fragments found deep in the cave; Great Hoary Bat (*Nycteris cinerea*), largest of the northern bats, possessing silvery grey fur and not normally a cave dweller, but wing bones were found in the cave, indicating they may have lived there at one time, and the Little California Bat (*Myotis californicus pallidus*), whose presence was indicated by a skull picked up amid the dust and debris of a low, inner room.

The chief resident of the caves for centuries has been the Mexican Free-tailed Bat (*Tadarida mexicana*), also sometimes called the guano bat because its species is especially noted for the huge quantities of commercially excellent guano it produces.

The Free-tailed Bat is distinguishable from the northern bats by its extended tail which reaches about an inch beyond the interleg membrane, by its short fur, and by a strong odor not found with other species. They also possess short, stubby ears, will weigh between 10 and 12 grams, and prefer caves more than most any other species.

The life of the bat is somewhat of a mystery to the average person, partly because they seclude themselves during the day in places relatively safe from discovery, and venture out at night when humans are not around and couldn't very well see them if they were.

Bats resemble the ordinary field mouse in appearance, except for their large, expansive wings. They are mostly insectivorous, living almost exclusively on insects they catch while in flight, chewing their prey into the finest bits with their sharp teeth. A bat will eat enormous quantities of food between sunset and sunrise, and on the basis of only two full meals a night will consume an amount almost half its own weight.

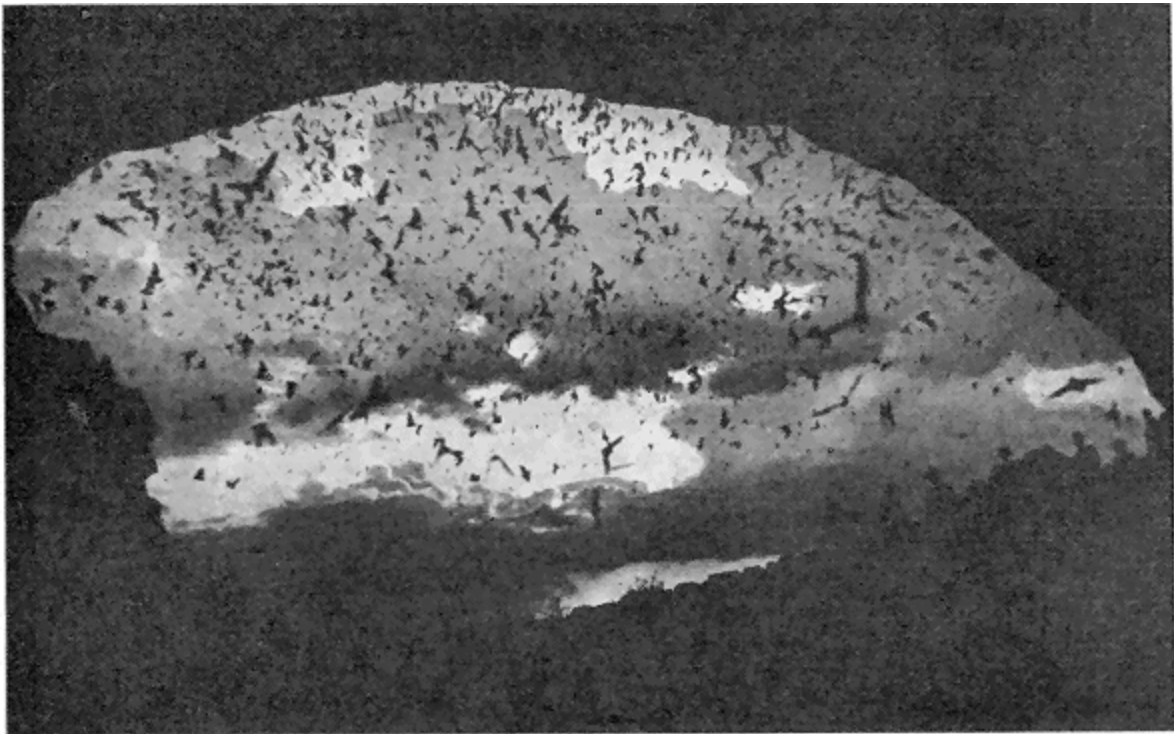
Their diet consists of beetles, flies, and other flying insects, most of which are enemies of civilization; thus the bat is economically a valuable animal.

They leave the mouth of the cave at dusk, heading first for a drink of water at the nearest available spot, then spend the night in search of their food. As the sun's rays begin to appear in the east they fly back to the cave where they dig their tiny claws into the ceiling and remain there sleeping until the sun sets in the west and the cycle starts all over again.

Due to their habits they have few enemies, the great horned owl being one. Very wide awake at night, he often catches an unsuspecting bat, chews the flesh, letting the bones drop to the ground where they remain as evidence of the owl's dinner.

The mating season is normally in March. Along about June the young are born, with this species only one to a female, although in some species a litter of four is born.

The single baby will weigh a fourth as much as its mother. Upon birth it clings to its mother's body night and day until it is strong enough to fly by itself, remaining tight even while the mother flies into the night in search of food.



© BY ROBERT NYMEYER

The famous bat flight—a nightly occurrence at the Caverns.

The habits of bats vary in respect to their living quarters. When insects are abundant, the size of the bat population increases, having been estimated at times to be as high as 7 million in the Carlsbad Caverns alone. As the supply of insects decreases, so does the number of bats. Where they go, and whether or not the same specific bats return in times of insect plenty is not definitely known.

One of the first surveys of the bats in the Caverns discovered that the number seems to grow in the fall, indicating that the cave was used primarily as a home principally for hibernation during the winter.

More recent observations indicate that, at the present time at least, the bats winter somewhere else and live in the cave only during the summer months, going elsewhere to spend the winter. Recently several thousand bats were banded by naturalists and of those which have been returned, one came from Jalisco, Mexico, some 800 miles south of the Caverns.

If both observations were correct, and it may be assumed that they were, for both were made by naturalists, then the habits of the bats do change, but

what prompts these changes is still a matter of speculation.

Other matters of speculation are just how the bat flies so accurately in the dark, darting swiftly past innumerable objects yet never hitting one of them. Scientists call this phenomenon "echolocation," a sort of sonar principle by which the little mammals let out a high frequency squeak and judge the distance of objects by the time required for the echo to bounce back to their sensitive ears.

Another mystery is how they find their way unerringly back to the cave, in the early morning as well as after a sojourn to some other area of the world, and how they know when the supply of food has increased to the point that it is sufficient for them to return.

Visitors to the cave often wonder how the numbers can be estimated. 7,000,000 is a lot of bats. This has been reduced to a relatively simple calculation. Cavern authorities counted the number sleeping in one square foot of space, found between 250 and 300 could squeeze into the area. The number of square feet of ceiling space occupied by the sleeping creatures was then multiplied by the bats per square foot, and the estimate was reached.

Life in the cave, it seems, has been governed by the bat, and even this little fellow can hardly be said to live there. Actually, he only sleeps there winters and during the day. He can't live in the cave all the time since there is no food there for him.

Are there any living creatures that dwell in the cave all the time? Only those that can find food in the cave, and since the eternal darkness prohibits photosynthesis which is vital to all growing plants, other animals or insects would find a food supply almost practically nonexistent, at least any form of plant food.

Yet there are cave crickets. One species was found which was entirely new to science, and they named it after the Caverns, calling the species *Ceutophilus carlsbadensis*. Their eyes do not seem to function, apparently depending upon their long antennæ to guide them.

But what do they eat? The contents of their stomachs have revealed bits of insect remains which they have scavenged from the piles of guano. Thus the cave crickets depend upon the bat for their existence.

The cave worm, actually an insect's larva, and a small spider also make their homes in the cave. How long these insects have lived in the cave is anybody's guess. Naturalists think the first few crickets may have fallen into the cave and remained there due to the abundance of the food supply, living peacefully and multiplying at leisure.

But this cricket's Shang-ri-la lost its comparative safety one day, maybe centuries ago, when a cave mouse is thought to have fallen in. He found the crickets abundant and tasty and, since he couldn't climb out, stayed in his new subterranean home. Maybe later Mrs. Cave Mouse met the same fate and together they became parents and grandparents of litters who have lived in the cave ever since.

The cliff mouse also lives his complete life cycle in the black depths of the Caverns, and together with the cave mouse they have a carefree existence, for none of their natural enemies live in the cave.

Any herbivorous animals which might have fallen into the cave would soon die, for the only vegetation to be found in the cave, aside from a certain amount of mold, exists in scant proportions near the cave entrances where there is some light and a little moisture. Thus only carnivorous or insectivorous animals or insects could exist in the vegetation-free depths.

Of the animals which occasionally frequent the cave, the cacomixle, commonly known as a ring-tailed cave cat or even just "ring-tail," is perhaps the best known. This slender raccoon-like animal doesn't spend all his time in the cave, for he is able to crawl in and out, but he does raise his family in the cracks and crevices of the walls, and eats an occasional cave mouse or guano bat. Outside he eats fruit and other small animals and insects. Apparently they have never been too numerous in the Caverns.

The only other animal to have made the cave its home is the great horned owl. His nests have been found in considerable numbers near the entrances where he has been living for countless years in comparative safety. But, like the bats, he ventures outside in search of food, and spends his nights away from the cave, so at best he, like the ringtail, is only a part-time resident.

So the cricket depends on the bats, the mice on the cricket, and the ringtail on the mice. Thus is the life cycle within the great cave.

But when did man first visit the Caverns? That question is indeed problematic and may never be definitely answered. The early Folsom people have been traced to an encampment site some fifteen miles south of Clovis, New Mexico, which they occupied some time between 10,000 and 15,000 years ago. Chipped stone dart points associated with a post-glacial species of bison have been found imbedded in the earth in rock strata going back that many years.

These wandering hunters were known as "Bison Nomads" and it is not at all improbable that they knew of the cave and frequented it. They could have been the first humans to have entered it.

Then, approximately 2,000 years ago, or at about the beginning of the Christian era, the Basket Maker Indians began to roam this southwest area of the continent. They may have been the physical descendants of the early hunters, but from a standpoint of culture they are credited with having brought agriculture from some focal point in Mexico much further south.

The Basket Maker Indians lived in this area for some eight centuries, and archaeologists mark three distinct periods of their civilization. The Basket Maker I people did much hunting and probably roamed the hills and valleys while the maize was growing, moving on to new areas after each harvest. The Basket Maker II group built simple dwellings, lived in caves, and grew numerous vegetables, including squash and beans. The Basket Maker III group established permanent villages, fired pottery, and elaborated on weavings established by both of the earlier groups.

The Basket Maker Indians are definitely known to have visited the cave, for their pictographs have been found on the entrance walls. What is more, skeletons have been found in the cave, some buried in baskets! These were found on shelves in the walls. Other skeletons were found buried beneath piles of guano far back in the cave, together with bits of pottery and broken arrow points.

As to how much the Indians used the cave remains speculative. Some archaeologists believe the cave was used mostly as a tomb, while others think the Indians may have lived in the cave for a certain length, basing this observation on the bits of pottery and other remains found along ledges within the walls. These, they point out, definitely were not buried with a body.

Most likely the cave was used for both purposes, for it served the Indian well as a place of refuge and defense, and it appears that the living quarters were located within a close radius of the entrance. Any more extensive use of the rest of the cave would have left more evidence, archæologists say.

But that was well over a thousand years ago. How about recent times?

Several authorities mention the fact that the existence of some form of bat cave or hibernation den was known to ranchers in the Carlsbad area as early as 1880-90. Great swarms of bats would rise out of a mound in the earth at dusk and go winging their way into the horizon. At sunrise they would return.

Was this the present Big Cave? Southern New Mexico contains numerous bat caves and many were known at this time. The existence of a new one didn't seem to overly excite the ranch hands, for exhaustive research has failed to reveal any positive knowledge of the existence of this particular cave as such prior to the time my father noticed the bats.

Millions of pounds of guano had been taken out of the various bat caves in this area by late 1900, and if anyone had stumbled onto this particular cave prior to my father's discovery, they would most certainly have immediately taken out a claim and proceeded to mine the guano, for there was so much of it that it assumed gold mine proportions.

The fact that my father was the first to take out a claim seems proof enough that he was the first to actually find the cave and, realizing its economic potential, did what any other person would have done, simply, take out a claim. It was the first ever recorded for this particular 20 acres and his claim may be seen today on pages 149 and 152 of Book I in the Eddy County Courthouse at Carlsbad.

3

The First Fifty Years

The discovery of the Carlsbad Caverns was not an earth-shaking event at the time. Its true value as a spectacle was to wait almost 20 years. Had it not been for the huge quantities of guano, my father and his friends might not have been especially interested in the cave, that is, sufficiently so to explore it further, since there were so many similar caves in this particular area of the State, and the discovery of one more didn't unduly excite anyone.

But because the guano looked like a worth while business proposition, the cave got more than casual consideration.

My father knew it was larger than some of the others in the area, and he gave it the title "The Big Cave." It was also referred to as "The Bat Cave" because of the millions of bats which would fly out of its mouth each evening, only to return in equal numbers the following dawn.

In those days it was primarily a guano producing cave and, except for occasional explorations by the guano miners and their friends, there were no extensive trips into the underground chambers by scientists. Only the commercial aspects of the fertilizer producing activities were recognized.

The guano was shipped to California distributors and was largely used in the fruit orchards there. The operation was not too profitable, as my father found out, and after he sold his equipment to Mr. Patterson the property changed hands many times. In 1906, for example, there were three different owners, and there was some splitting of ownership whereby an owner would sell a part or half his interest.

Also, there were other sections which were owned by independent operators. Prior to 1906 the Santa Fe Railroad owned 40 acres over the east portion of the cave.

By around 1912 the bulk of the guano had been removed and activity around the cave ceased. There were little spurts of renewed interest from

time to time, but there was actually little news about the cave for several years.

In later years people have often speculated as to whether or not the great Carlsbad Caverns would ever have been discovered had it not been for the attention caused by either my father's discovery quite by chance, or from the dark clouds of bats which emanated from the cave each evening.

We know now that even if those two events had never occurred—even if there were no opening on the surface of the earth, if there were no habitation of bats, the great cave would have indeed been discovered.

And historians now like to think that the cave was actually discovered twice. The first, as we know, was when my father stumbled upon it quite by accident back in 1903. This was the discovery of the guano beds, the beauty and grandeur of the Caverns being secondary at that time.

Then came what historians have since called "the rediscovery!"

And it, like the first discovery almost 20 years before, came about quite by accident!

At the end of, and immediately following, the first World War, around 1919 and 1920, the United States Government Commission of Reclamation was constructing several dams along the Pecos River some 20 miles east of the site of the big Bat Cave. The purpose of the dams was to impound river water for irrigation of the surrounding agricultural lands.

But for some strange reason the dams didn't seem to be doing any good. The water seemed to disappear underground. The Commission felt it was useless to consider the construction of any more dams in the area until the mystery could be cleared up.

They took the matter up with the U. S. Geological Survey, and again the question was unanswered. It seemed strange that the water could not be held and used to irrigate the surrounding farm lands.

The U. S. Geological Survey said they felt the problem was of sufficient value to merit a complete investigation. They would, they said, send a party to the site to see if they could unravel the riddle of the river's strange behavior. They advised that no more dams be constructed until the source of the trouble could be determined.

The assignment was given to Dr. Willis T. Lee.

He pondered the situation and knew from experience that there were several possible solutions, yet he couldn't do much more than hazard a guess as to what the trouble might be. He also knew that when nature is concerned he might stumble onto an entirely new reason.

Before he left Washington, Dr. Lee had reviewed all the available information on the geological formation of the earth in the area and was quite well informed on its composition.

Now, on the land itself, he began to study the situation from every angle. He found the limestone strata throughout the entire area were a bit different from that which he had previously seen in other parts of the world.

Gypsum and salt are soluble in water, and when the dams were built they obviously raised the water level several feet above its natural contours or limitations. The water would then find its way into these alternate layers and would seek lower levels, dissolving the rock salt and gypsum as it did so. With nothing to hold it behind the dams, the water would disappear into the earth and reappear again many miles below the spots where the dams had been built.

While Dr. Lee was in the vicinity, he decided to have a look at the famous Caverns of which he had been hearing so much, and was promptly invited to see these underground wonders since he was so close to them. He knew of the existence of the many bat caves or guano caves in the southeastern part of New Mexico, but had not suspected they contained any real beauty. Now, for the first time, he was to learn otherwise.

Prior to this the United States Land Office had sent Mr. Robert Holley to survey the Caverns, and by the time Lee visited these underground chambers, Holly had surveyed some three and a half miles of their interior. It was Holly's report which brought about the designation of the area as the Carlsbad Caverns National Monument.

Dr. Lee was enthusiastic in his description of the great Caverns as a thing of immense size and beauty. Here was something that had been kept a secret long enough, he felt, and the government should do something about them.

His report was the spark that set off a chain reaction. Now, for the first time, national magazines began to publish reports of this huge, fabulous cave in

southeast New Mexico. In the latter part of 1923 several articles appeared based on Dr. Lee's descriptions of the underground wonders. Now, at last, the world was learning of the existence of this fairyland of unmatched scope anywhere in this country, perhaps in the world.

But prior to this time the National Geographic Society had become interested in Willis Lee's findings. Certainly this was worthy of the Society's attention. Accordingly, they set up a grant of \$16,000 to finance the first full scale exploration of the Caverns. They selected Dr. Lee to head the party.

Dr. Lee's mission this time was much different. He wasn't concerned where the underground waters were going. Rather he was set upon finding and reporting the scenery about which he knew the world would want to hear.

His subsequent report in the National Geographic Magazine (January, 1924) marked the first full scale description of the Caverns, and for the first time the public saw a series of pictures of the beauty of this underground wonderland and read something about the various rooms, their gigantic size, and the extent of the formation.

Dr. Lee's party didn't find a smooth paved road from the town of Carlsbad to the site of the Caverns. By contrast, the 28 miles of hilly, bumpy, dusty road were covered by automobile in about two hours. The road had been used primarily by the fertilizer companies in transporting the sacks of guano to Carlsbad for shipment to California.

At this time there were several dwellings at the entrance to the cave, an engine house and two hoisting shafts.

The exploring party engaged Jim White as a guide to accompany them below. At this time Jim White probably knew the Caverns better than anyone else, for he had lived in the vicinity of the caves for more than a dozen years and had served as a guide for many persons who had made trips below to behold the beauties to be found there.

Jim had talked of the cave and its marvels to anyone who would listen, and for years had tried to stir up some enthusiasm for the wonders that lay beneath. He had explored much of the cave and could talk intelligently about it. Obviously, he was the best guide that Dr. Lee could have found.

The descent at that time was not a very glamorous undertaking. Dr. Lee's party entered, two at a time, by standing in an old iron guano bucket and letting the "elevator" engine slowly lower them more than 150 feet below the surface. The wire rope fastened to the bucket ran over a pulley at the top of the derrick and was manipulated by the engine. Dr. Lee was apprehensive of the contraption, but it never failed his party.

In that early and historical excursion into the great Caverns, Dr. Lee noted the weird, eerie effect the shadowy forms, high ceilings, and irregular floor and walls had on his group. They found it difficult to believe what they were seeing. Below, the opening above them became but a speck of light.

They noted the great distance from wall to wall, the many alcoves, the roughness of some walls, the polished smoothness of others.

Some distance from the foot of the shaft they came upon a large pit some 150 feet deep which they called Yeitso's Den. The name was later changed to the Devil's Den.

This part of the cave was much as the surging waters of the past several million years had left it. Travel was slow and dangerous. Steep walls had to be climbed and it was necessary to cut steps in the stone. The forward progress was indeed tedious in those early days.

Dr. Lee's report to the National Geographic Society was filled with descriptions of the colorful stalagmites and stalactites, the beautiful formations of flowstone and dripstone, and the subject matter to be seen in the formations all around them. He was equally impressed by the "chambers of unbelievable dimensions," some of which were "several hundred feet wide" and ceilings "so high our lights won't touch them."

He measured the smallest of three rooms which he entered after his journey down the main corridor, which he describes as "slow and laborious." The room measured 160 by 140 feet. "The middle room was three times this size, and the first room was even larger," he said.

He noted the "curtains" formed of onyx created by the deposit of lime carbonate dripping from above. The ceilings were found to be covered with curtains of these stalactites which had grown together during the ages. At times they met with stalagmites to form solid pillars of seemingly carved formations.

At times the floor was rough and covered with fallen blocks of stone. In other places it was quite smooth and flat, obviously the result of years of solutive action by water. Stalagmites do not grow in areas covered for any length of time by water, such as an underground pool. In one place the stalactites were so slender and numerous that Dr. Lee referred to them as resembling "a porcupine in posture of defense."

Dr. Lee's group gave names to many of the rooms and formations, many of which have since been changed. But one which has stuck with all explorers, both before and since Dr. Lee's excursion, is the Big Room. This underground colossus impresses everyone who sees it, and Dr. Lee hesitated to estimate its dimension. "Over half a mile long," he said at that time, "with a ceiling some 200 feet high."

Big as the room was, he felt it was equally remarkable for its numerous dripstone decorations, their variety and size. Some, he noted, rose to 50 feet in height.

He found some stalagmites were still "growing," for they were darker in color than those no longer alive, indicating the presence of water and continued growth. As the supply of water ceases they become lighter in color, almost powdery.

One of the largest he described were the Twin Domes, which he said were more than 100 feet high and some 200 feet across at the base. They are, today, much the same as he first observed them and are one of the favorite formations of all cave visitors.

In the light of Dr. Lee's primitive torchlight, the base showed up a dark green, with the grooved pillars light cream in color, both glittering and sparkling as the light struck them.

Other interesting observations by the party included the location of extinct seeps on the floor of the big room, the basins of which were up to about 50 feet in diameter, pedestals of onyx resembling toadstools and so called by the party, and an underground series of chambers through one of which flows a stream of clear water.

Dr. Lee's party was unprepared for the vastness and extent of the Caverns. Although his group had come prepared for a sizeable task, the job

overwhelmed them. Although they mapped much of the underground passageways and rooms, they could see much was left undone.

But what they had done was enough to prompt the United States Government in declaring the site a National Monument. On October 30, 1923, Secretary Work announced that President Coolidge had proclaimed the area a National Monument.

Now it was official. If the government proclaimed it as a National Monument, it must be worth a visit. So thought many people throughout the land as they read about it in the many national magazines that ran descriptions of the Caverns based on Dr. Lee's findings.

My father's original holdings consisted of 20 acres. Now that the government had taken over, the size was increased to 700 acres.

Spurred by Dr. Lee's first exhaustive article in the National Geographic in January, 1924, people began to travel to this underground land of wonders. Diplomats, governors, people of all walks of life wanted a first-hand glimpse of what had been found.

There were no adequate facilities with which to accommodate them. Silk hats in guano buckets didn't look exactly right. Improvements were in order.

Dr. Lee reported how preliminary his work had been. More exploration was needed, and as a result he returned to the Caverns in the spring of 1924 in order to further extend the work he had started when he first saw them.

Exploration began the middle of March and lasted for six months, during the heat of the summer when it was as high as 115 in the shade—and, as Dr. Lee says, there was no shade. Yet inside the cave it remained about 56 degrees, which is constant the year around. During this time the party played host to the Governor of New Mexico, the Governor of Texas, the Director of National Parks, and many other high ranking dignitaries.

Dr. Lee and his party set up their camp at the mouth of the Big Cave, using tents and the rough board shacks left by the fertilizer company years before. Their closest source of water was a spring about a mile away and their supply had to be carted by horse or burro.

While working in the cave the members of the party used kerosene torches, but upon discovering that these gave off more smoke than light they

abandoned them in favor of gasoline lanterns.

In order to safeguard themselves, they would unravel balls of twine as they explored remote sections and passageways. In this way they could always find their way back to safety. Some of the members of the party were hesitant to wander far from the well-worn trails, apparently afraid of the unknown and admitting their weakness. Others allowed their curiosity to dominate their fears and were continually searching for new finds.

The party obtained its supplies from Carlsbad, about 28 miles away, and during the summer months the fruits and vegetables were kept in a special "refrigerator" in the cave where the cool, even temperature was put to constructive use.

The party's first task was to improve the various trails. Large rocks were pushed aside and, where this was not possible, safe ladders were constructed when the obstructions could not be circumvented.

On Dr. Lee's previous trip he did not explore the many small chambers and rooms leading off of the main halls and central areas. This time he made these deviations his specialty. Numerous nooks and corners were explored and mapped. Only those that for reason defied access were left alone and kept their secrets sealed within.

The entire party was amazed at the ornate beauty they found all around them. Many of the rooms were being seen by scientists for the first time, who would subsequently describe these colorful wonders so that all the world would know.

Of the Dome Room he later wrote "from the arched vault hang pendants of ornate character and on the wall are sheets of delicately colored onyx resembling portieres looped back in graceful folds. One is impelled to touch these folds before he is convinced that they are cold, hard stone."

In another chamber Dr. Lee noted how some of the stalactites would ring when struck lightly with a stone or metal instrument. One that was broken revealed a small trickle of water within. It was, said Dr. Lee, excellent drinking water.

Further on the party found a large hole which was 100 yards across and almost one third that deep. Onyx marble was to be seen all around, on the

walls and the ceiling which was some 200 feet above the floor of this indentation.

Climbing down a wire ladder, Dr. Lee and his associates found several "fountain basins" many of which had a crust of onyx which resembled somewhat the ice that forms over a small body of water in the winter time, after which the water recedes leaving the thin sheet of ice suspended in air.

Other unusual formations were the many pisolites, small, spherical bodies of hard stone which jewelers call "cave pearls." Some were immovable while others were loose, probably never having been able to fasten themselves due to the continuous agitation caused either by a flow of water underneath them or from water above dripping down around them. These "cave pearls" grow from a tiny beginning through hundreds of years by the constant deposition of minute quantities of calcium carbonate left by the dripping or flowing water. Those still active are moist and glistening. The dead ones are pale, with a powdery, dull look. Because of the fact that these "pearls" resembled eggs in nests of stone, the room was called the Rookery, a name which has remained ever since.

In places where water drips at just the right frequency, the water will splash to the surrounding walls, leaving as it dries there its small particle of calcium carbonate. In places it hardens to form funnel shaped vases. In other places it dries too rapidly, leaving a soft powdery accumulation.

Still another formation was the "helictites" which are certain types of stalactites which didn't know which way to go. They appear to defy the laws of gravity, for they may start in one direction and then dart off in another direction. Such formations were caused by certain types of crystallization. Some chemicals will "grow" upward as they dry and expand, and this is probably one explanation, although it may not apply in every case.

By the early fall in 1924, when Dr. Lee and his party had completed this second thorough study of the Caverns, he was most enthusiastic about what he had seen. His observation at that time describes the underground wonders exceptionally well. Telling about the exploration in September, 1925, issue of the National Geographic, he said:

"The Carlsbad Cavern, New Mexico, is the most spectacular of underground wonders in America! For spacious chambers, for variety and beauty of multitudinous natural decorations, and for general scenic quality, it is king of its kind."

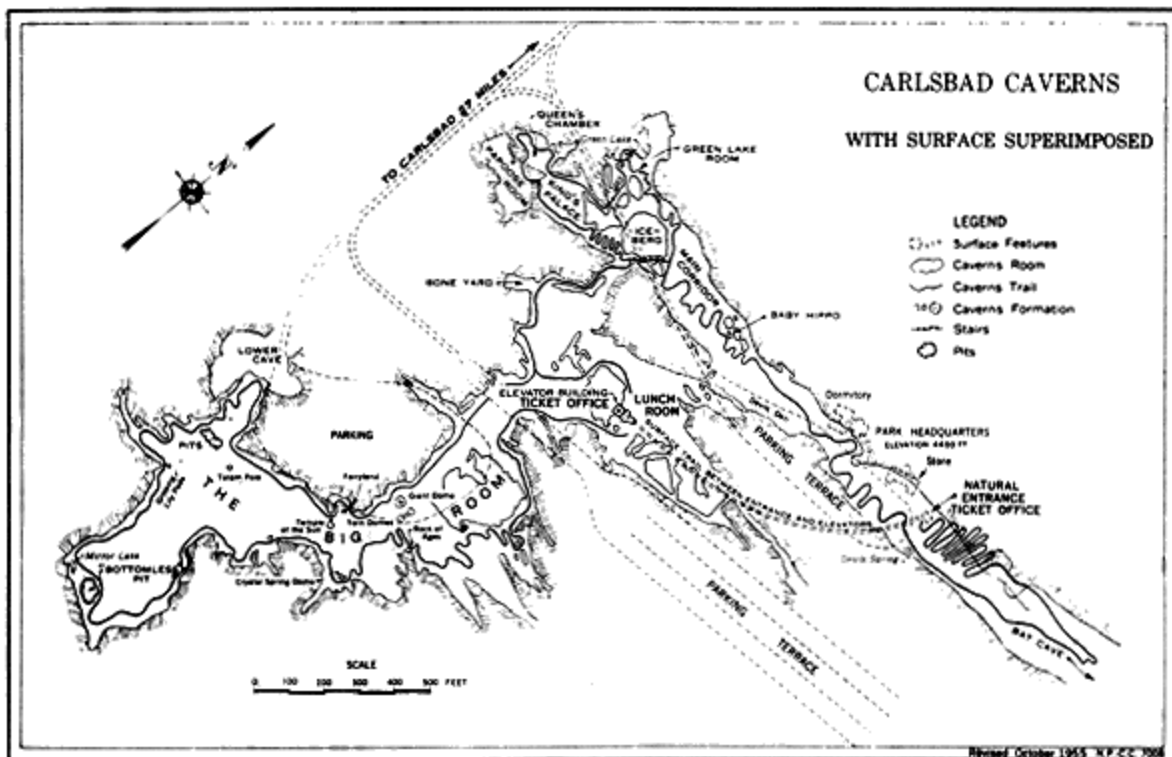
The party had covered a total of some 23 miles, which they had patiently mapped and surveyed, photographed, and described. The "rediscovery" had now been completed—and in grand style. The next descent into the great cave was to be by the public itself, and in ever-increasing numbers.

In order to meet this onrush of sightseers, improvement of the interior of the Caverns and the approach from above ground was deemed necessary. Now that the area was a National Monument, the Government gave its attention to the area. Because of his great interest in, and knowledge of the Caverns, Dr. Lee was appointed the first custodian. He served without pay. Jim White became an unofficial guide, later being appointed a park ranger by the National Park Service, finally becoming chief ranger.

Now, with some system of procedure mapped out, improvements began in earnest. All of the major trails were greatly enlarged and smoothed over so that their passage by the public would be safe. Lighting of the halls and chambers was also begun, although most of the illumination was furnished by the lanterns carried by the guides and members of touring parties.

In 1927 the Government could see the Caverns were beginning to take hold as a show place, a vacation mecca, and additional help would be required. They sought a man who could take complete charge for the Government. Someone suggested a man who was at that time in charge of the volcanoes on the Hawaiian Islands, Colonel Thomas Boles. Here, they felt, was a man who could do what was necessary at the great Caverns, and as a result Colonel Boles was appointed the first Superintendent in that year.

During the cave trips at this time everyone carried a box lunch, for a full scale trip below required the best part of a day. With the increasing attendance the government decided an underground lunchroom would be an added convenience for the visitors.



[Click on map to view larger sized image.](#)

Accordingly, in 1927, a suitable room over 700 feet below was selected for this purpose, one of the guiding factors being that it was located approximately at the point where the visitors passed at noon-day.

Picnic tables were set up and box lunches provided cafeteria style to the hungry travelers whose appetites were heightened by the brisk journey in the ever-cool atmosphere.

As interest in the Caverns continued to increase, the Government realized the site was worthy of more than simply a monument rating. The matter was taken up before Congress two years later, and on May 14, 1930, Congress voted to make the area a National Park.

It is also significant that it was at this time that the official name became the Carlsbad Caverns National Park, the big caves being named for the town of the same name some 28 miles northeast and which has always served as the chief supply point for the vicinity.

The surface area was also enlarged at this time and was extended to include several more acres. On February 21, 1933, it was enlarged to include more

than 10,000 acres.

It was at this time that the greatest improvement yet to be made at the Caverns was undertaken. This was the installation of the elevator, which was completed in 1932. The lift is 754 feet and shortens the walk-in trip by about one and three-quarter hours. The elevator lets its passengers out at a point adjacent to the Big Room and the lunchroom. Installation of the elevator made it possible for elderly people and those who were not physically capable of walking in through the natural tunnel corridor to view the magic wonders of the great subterranean fairyland. As soon as it was completed, a great many more people came to see the great Caverns.

Walking into the Caverns is a thrill within itself and, since the trail is downhill, few people find it much of a chore. Going out is something else again, and the great majority of people who walk in are glad for the opportunity of taking the elevator up to the surface 754 feet above them.

In 1939 the Park was extended to include some 49,000 acres, and within this area are 30 other caverns, some of which have been quite extensively explored while others are relatively unknown. In the main, most of them are similar, though none begin to compare with the Carlsbad Caverns in size, and the details and formations often differ as well.

Three years later the Government purchased the east wing from T. A. Blakely. This was the section lying for the most part over the guano beds which in previous years had been worked so extensively. The purchase of this last segment put the entire known area of the great cave in the hands of the government.

In 1946, Colonel Boles, the park's first superintendent, who had served through 19 pioneering years, was transferred to Hot Springs National Park. By that time he had spoken to well over 2 million visitors, many of whom were world famous. He says he has made over 5,000 complete trips through the Caverns and on each "I saw something I missed before." Much of the early progress was due directly to Colonel Boles' leadership and foresight. Today, having retired from the National Park Service, he makes his home in Carlsbad, where he serves as public relations counsel for the Potash Company of America.

Donald S. Libbey followed Colonel Boles as Superintendent of the Caverns and he in turn was succeeded by R. Taylor Hoskins. During the first eight years of his office, Supt. Hoskins says almost 3,000,000 people paid to see the limestone creations which many have tabbed "The Eighth Wonder of the World." Those who make the underground tour come away feeling that appellation can't be far from right.

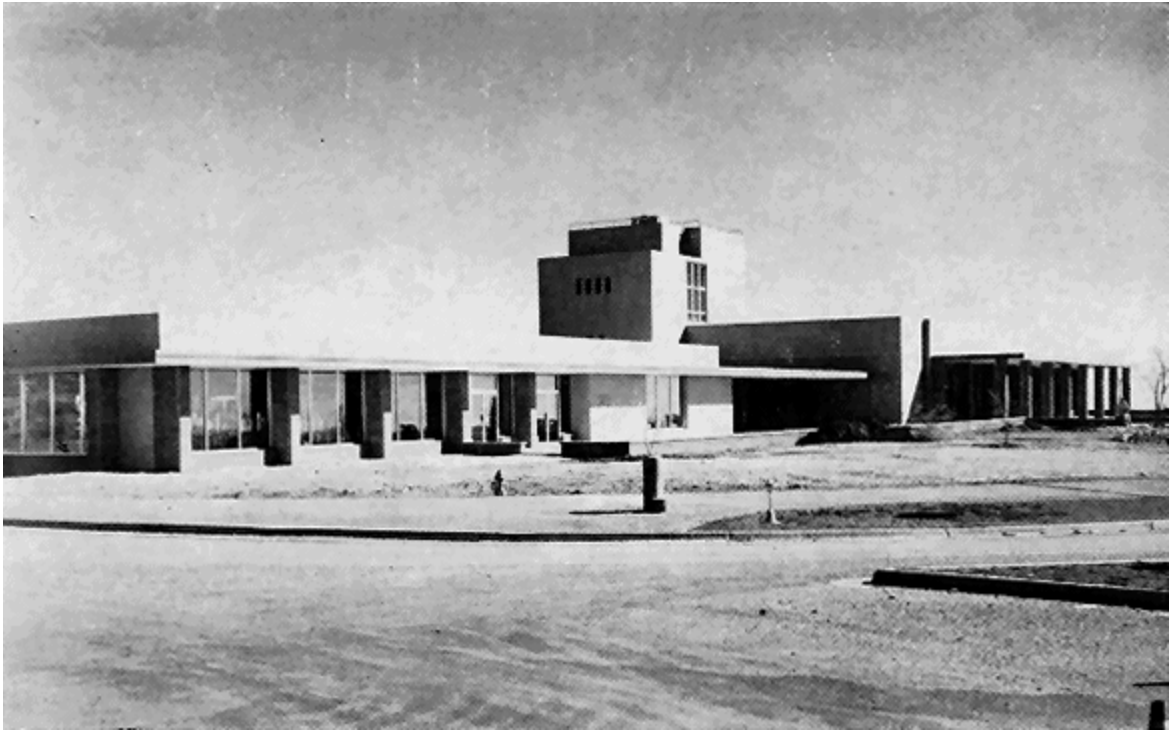
Though the government spent some money years ago in improving the Caverns and preparing them for visitors to see in safety, the returns on that investment are now counted. Although the park receives somewhat over a quarter of a million dollars appropriation from the federal government each year, it returns almost twice that amount, the sum being obtained from tourist fees and similar income.

Yes, these first fifty years have indeed seen many changes within the Caverns. Nature's work has been protected and left intact, but the lights, the lunchroom, the elevator, the improved trails, and the many other improvements have been made by man for man's enjoyment of what the ages have bestowed—what the centuries have given him to see and enjoy.

Already people from the four corners of the globe have beaten a track to the Caverns' door, to behold in amazement and wonder and awe. Many millions more will come in the years ahead, and their children in the years after that. Here, some 700 and 1000 feet below the surface of the earth they will marvel at the work of the centuries—that began some 60,000,000 years ago.

Part III
THE BIG CAVE TODAY

By JOE N. LONG



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Visitor's Center Building—Completed in 1958

1

From Above

When the traveler finally decides to include a visit to the Carlsbad Caverns in his itinerary, or perhaps make a trip to the Caverns an end in itself, he procures a map of the Southwestern United States and searches for the little town of Carlsbad. Far down in the southeast corner of the state of New Mexico he will find this colorful little community, resting serenely some 40 miles north of the Texas—New Mexico line, about a hundred and sixty miles east of El Paso, and about a hundred miles northeast of the Mexican border.

Carlsbad is now a thriving little city of approximately 20,000 people, situated on the banks of the Pecos River at an altitude of slightly over 3100 feet. It is often referred to as the potash capital of America, and the growth of the potash mines has had a direct influence on the increase in population, which has been quite rapid in the past few years.

Carlsbad was settled in 1888 and at that time was known as Eddy. In 1906 the Government purchased what remained of a private irrigation system which had been destroyed by a flood of the Pecos River two years before, and developed it into the Carlsbad Reclamation Project. This assured adequate irrigation to the surrounding fertile lands and reduced to a great degree the danger from subsequent floods. As a result, agriculture flourished and has continued to be of prime importance in and near the area. The soil is especially adapted to the growing of alfalfa and cotton.



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Air view of the entrance and surrounding area.

The visitor can reach Carlsbad by air, rail, or automobile. Continental Airlines provides daily airplane service to the Carlsbad Airport, and connections can be made with nearby cities on all of the major airlines flying through this section of the country.

The Santa Fe Railroad operates regularly scheduled trains to Carlsbad from Clovis, connecting with transcontinental trains to Clovis from coast to coast. The Southern Pacific connects at El Paso, as does the Texas & Pacific, and the Rock Island.

But there is little doubt that the great majority of visitors come via the family automobile. From U. S. Highways 60 and 380 to the north the motorist turns south on Highway 285 through Roswell and Artesia to Carlsbad. Here Highways 62 and 180 intersect, going from Carlsbad to Whites City, 20 miles south of Carlsbad, where a turn-off on State Highway 7 takes the traveler seven miles to the Caverns' entrance.

Airplane passengers and those coming by train will take the same motor route from Carlsbad, since it is direct, and the only route from that city.

Special busses known as the Carlsbad Caverns Coaches make the trip at frequent intervals, and also operate from El Paso, which is about 145 miles away.

As the Caverns-bound sightseer heads toward his objective from Carlsbad or El Paso, he will find himself in the heart of the desert. There is hardly a tree that is more than head-high. Visitors seeing the desert for the first time will marvel at the yucca and sotol along with other less majestic desert vegetation. The stately yucca points its bloom skyward and is a thing of beauty in the spring. Among its grass-like leaves are often found mockingbirds, thrashers or cactus wrens, who like to build their nests amid the security afforded by the spine-tipped bunching of the limbs and branches. The uppermost point of the sotol sometimes reaches as high as 15 feet in the air, while the yucca, often referred to as the Spanish bayonet, occasionally looms to 25 feet. Indians used its leaves for baskets and its roots for soap, and the sotol served the Indian as food, both cooked and raw, the crisp rich centers tasting much like celery hearts.



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Cavern Entrance

There are several other species of cactus, and to the newcomer these will be a new experience. Besides the Devil's Head and the branching cane, one sees the thousand-headed cliff cactus, the large leaf "prickly pear", various kinds of mescal, and numerous smaller varieties. The octopus-like ocotillo, spreading its fingers outward to the sky, is numerous, picturesque, and unforgettable.

The traveler who is fortunate enough to make the trip during the spring months will be extra rewarded by seeing this desert flora in bloom. Every color of the rainbow will be noticed, but primarily the desert blooms of cacti are for the most part quite pronounced in the lighter, soft colors, especially in the light pinks and yellows. A few give forth edible fruit, and many are known for their ability to hold moisture, often a lifesaver to man and beast.

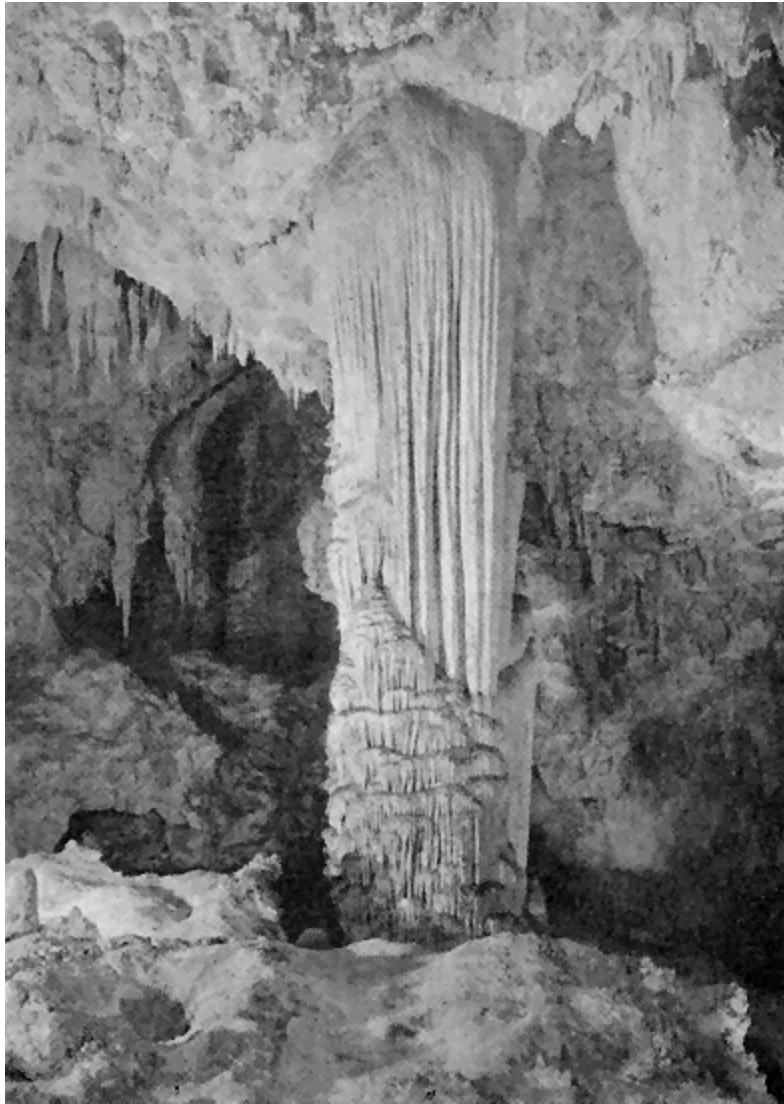
What trees there are will be found mostly along the banks of, or adjacent to, small streams or rivers which supply them with the water their roots must have. Besides several species of oak are the black walnut, black cherry, New Mexico maple, junipers, Mexican buckeye, the small green ash, an occasional pine struggling for existence, and numerous mesquites. There are also several nut and berry trees which have long been helpful in preserving the wild life that inhabits the area.

Desert wild flowers accentuate the land in spring with their dots of red, yellow and blue. In places they are so thick as to resemble a crazy quilt of many hues and shades.

Although the area gives the impression of a barren wasteland to the inhabitants of less arid regions, there are many forms of wild life which for countless centuries have made the area their home. Although the Red man's arrows and the white man's bullets have reduced the herds of wild game which once roamed the hills and valleys, there are still a sizeable number which may occasionally be seen.

Among these are the gray mule deer, still found in the narrow gulches and on top of rocky ridges, jack rabbits which continue to multiply and dart rapidly hither and thither, being most noticeable at dusk, rock squirrels which dwell primarily in the rocky cliffs, and numerous kinds of rats which seem to have little trouble eking out an existence.

Occasionally an antelope is seen, but the great herds of buffalo are now almost entirely gone, as are the mountain sheep, the Merriam elk, the bear and wild turkey. The latter two were sacred to the Red man and were protected during his inhabitation of the Western United States, but the coming of the white man left no bird nor beast in safety.



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Statue—Green Lake Room

Along the banks of the nearby Pecos River one may see geese and ducks, and occasionally one can find a crane, a heron and perhaps a soft-shelled turtle. Scaled quail are in the vicinity, as are Mearns or fool quail. Buzzards,

several varieties of owls, swifts and flickers are a few of the many wild birds which dot the countryside.

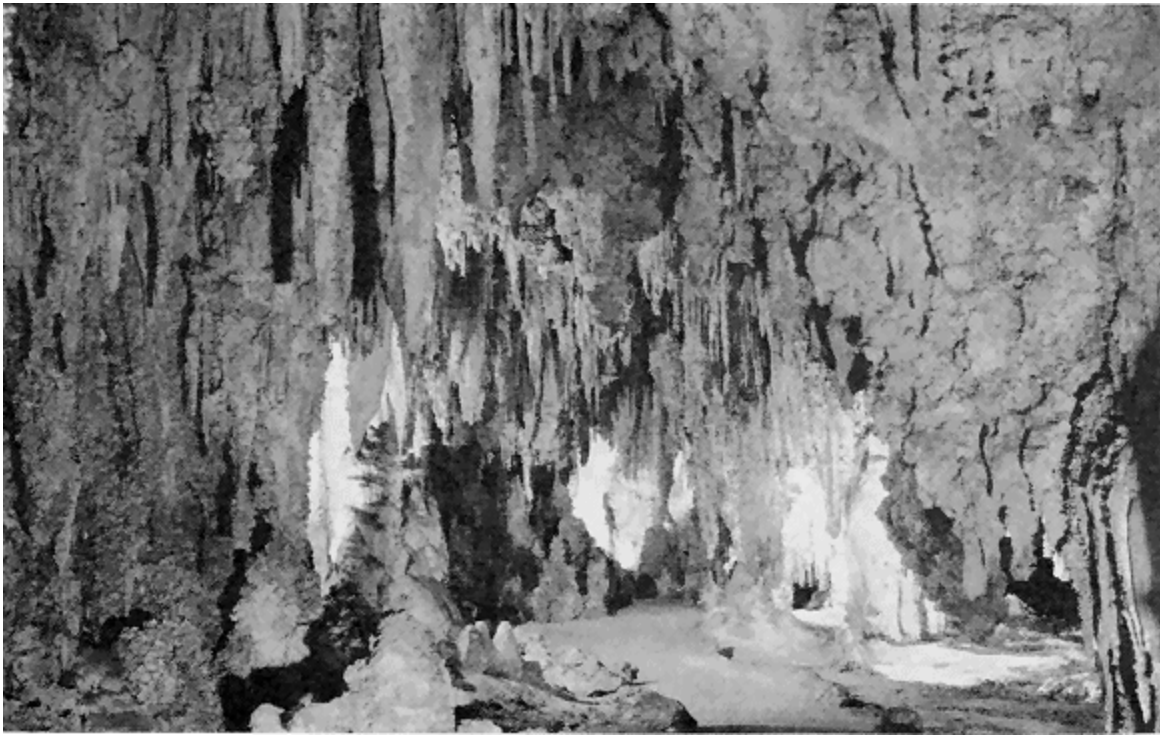
On the ground there are several kinds of snakes and lizards. There are several varieties of non-poisonous snakes such as the bull snake, the Mexican black snake, the coachwhip, the ring-neck, and several species of water snakes. Three poisonous snakes are found, the black-tailed or "Texas" rattler, the prairie rattlesnake, and the large western diamond-back.

Lizards are quite numerous and will be seen by any traveler who keeps his eye peeled along the roadside from which they suddenly dart, running swiftly to the other side of the highway. There are several species which consist of the rough, scaly rock lizard, the smooth, slender whip-tail, the brilliantly colored collared lizard, and the bar-tailed lizard. There are several smaller species including the horned lizard, often referred to as a horned toad. Scorpions and spiders are numerous, and there are many other smaller insects that inhabit the area and serve as a source of food for the birds, reptiles, and small animals.

As the journey to the park continues the visitor passes by small adobe shacks which may serve some cowboy or sheepherder as the necessary shelter while tending his herds. Range cattle are numerous and occasionally several goats will be seen. An old prospector or "desert rat" who can't stand the hustle of city life may perchance be found out on the dry, arid wasteland hoping for a find within the hills or neighboring Guadalupe Mountains which by now are quite plainly visible to the west.

Stretches of the land are alkaline, white as though touched by the fingers of Jack Frost, and now and then a dry lake reveals only the sodium given up as the water evaporated and left its soft white lining.

Beyond, the Guadalupe loom up a deep purple on the horizon, darkened by the foliage of pine trees that populate the peaks of this mountain range which reaches a height of almost 9,000 feet. It is the first mountain range to be encountered by the traveler coming from the east after having journeyed across the flat plains of Texas and the other midwestern states.



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Along the trail in the King's Palace

For the visitor who is coming from El Paso, the Guadalupe Range must be crossed, and the pass is located next to Guadalupe Peak, the highway reaching an altitude of 5,288 feet at this point. Guadalupe Peak is the highest mountain in the Guadalupe Range and it is also the highest point in the state of Texas.

It was near this spot that the Butterfield Stage passed on its way from St. Louis to San Francisco. The famous trail passed the southern tip of the Guadalupe near Guadalupe Peak, coming this far south in order to avoid the snow covered Rockies which made winter travel impossible and summer travel difficult. In 1857 the Butterfield Stage brought mail to the little settlements in southern New Mexico, giving the people there a more frequent contact with the rest of the world. The Butterfield Line was discontinued four years later, yet the history of its brief existence lives on in the minds of New Mexico residents today.

The Guadalupe foothills are covered with shrubs and hardy vegetation, growing above the limestone, shale and gypsum below which are harbored

numerous caves similar to the Carlsbad Caverns. There is Deep Cave, appropriately named by Carl Livingstone, its discoverer, and Slaughter Cave, also known as New Cave, which is one of the best known. It has been explored and contains many beautiful and unique formations. Some of the cavern sequences in the motion picture "King Solomon's Mines" were filmed in New Cave. Difficult access to its entrance keeps it closed to the public.

McKitterick Cave is located near McKitterick Springs, some 18 miles west of Carlsbad, and holds fond memories for many of Carlsbad's older residents. They recall how, as long ago as 1885, the gay young blades would take their ladies fair on a trip to the cave as a form of amusement, leaving early in the morning and packing a mid-day lunch.

Others, not as famous, are Hidden Cave and Cottonwood Cave—part of more than thirty in the area well enough known to have names, and only a few of more than a hundred smaller caves which perforate the foothills in the area. Many of these are located within the park boundaries.



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Under the Guillotine—King's Palace

At the edge of the boundary, the traveler arrives at the turn-off point to the Caverns' entrance. If the journey is made in winter, zero temperatures can be expected at the coldest times. In summer the desert heat may reach 100 and often does. Winter or summer, the route is open, for the Caverns never close, being open to the public the year 'round. In winter the sandy soil may be buried under a blanket of freshly fallen snow, the air with a light snap and crystal clear 'neath a warm winter sun. In the summer months a bluish desert haze often filters down on the surrounding countryside, but it does not stop even the slightest breeze from whipping up a soft funnel of dust from the dry, parched earth, or sending a tumbleweed rolling along a haphazard path towards an undetermined destination.

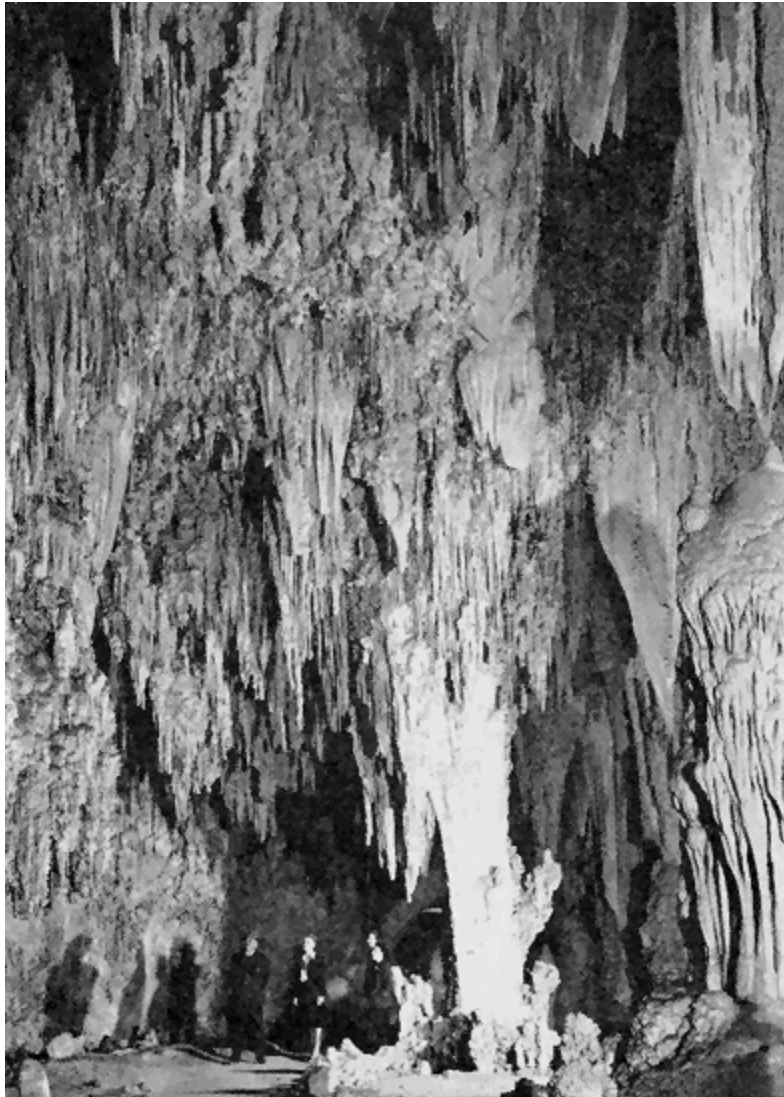
But the traveler's destination by this time has almost arrived. Highway No. 7 leads seven miles up Walnut Canyon road through rocky mesas covered with choice examples of desert flora, and well they might be nice, for now, inside the park boundary, park laws protect their safety and no one is allowed to touch them. They live their normal lives secure and aloof from man's sometimes destructive inclinations.

The visitor is now within the more than 45,000 acres that constitute Carlsbad Caverns National Park. The paved highway that leads to the Caverns' entrance is indeed a decided contrast to the wagon wheel trail followed by Dr. Willis Lee and his party of explorers which first covered the route by automobile in 1923. His engine was overheated and steaming after taking two hours to make the trip from Carlsbad.

Now, at last, at the entrance to the famous Caverns, the visitor is excited at the thought of what he is to see, but few anticipate the real treat that is in store for them underground.

Here, at an altitude of 4,350 feet above sea level, is found the natural entrance to the cave. Park regulations are studied in order that no bounds may be overstepped. No living thing may be touched or disturbed, thus assuring protection and perpetuation of plant and animal life within the park. The walls of the Caverns may not be touched, defaced, scratched or marred in any way. Canes or walking sticks may not be carried except upon permission of a park official, who will decide if such cane or stick is necessary for the owner to make the Caverns trip.

No dogs are allowed in the Caverns. They may be kept in the kennels provided for the purpose at a nominal charge. Cameras may be used on the surface, but within the Caverns they may be used only on the special photographic tour, full information concerning which may be obtained from the superintendent's office.



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King's Palace

There are no accommodations for camping or overnight lodging within the park, but motels are available at Carlsbad and smaller communities along the highway in either direction from the Number 7 turn-off.

Tiny children will not appreciate the beauties of the Caverns and their lack of interest will detract from full enjoyment of the trip by their parents and older members of the family. Accordingly, the park service maintains a day nursery where children may stay during the trip of their families below ground. The charge is \$1.50 per day, which includes lunch.

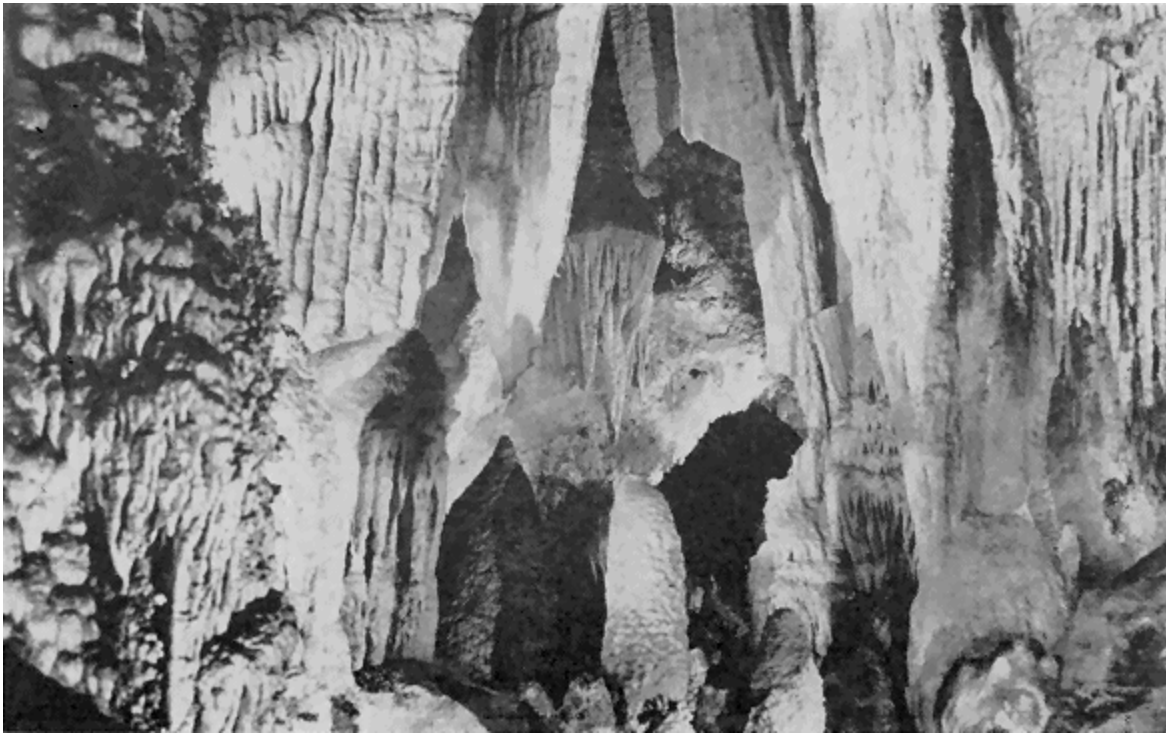
Before going below the visitor should make certain his clothing is adequate. The temperature in the Caverns remains at a constant 56 degrees the year around. The winter visitor is usually well equipped with proper clothing, but the summer visitor, dressed for the hot summer temperatures above ground, may become quite chilled after a few moments in the sudden change of some 30 to 40 degrees. Consequently, warm clothing should be worn for the Caverns trip, which should include a medium weight coat or sweater plus heavy skirt or trousers. High heeled shoes are not recommended for the journey, and a change to the low heeled variety is strongly suggested.

The trip through the Caverns requires approximately four hours, and a box lunch may be taken if desired, but the added inconvenience it imposes is hardly worth the trouble since a large lunchroom with a capacity for serving 1,200 people per hour has been established in one of the rooms below and each tour stops 40 minutes for lunch at this spot.

And now, armed with all of the essential information concerning the park, the Caverns, and the regulations concerning the forthcoming visit to Nature's fairyland, the visitor is eager to get on his way.

Park rules prohibit anyone entering alone or without a guide. Trips are conducted every day in the year, beginning hourly in the summertime when the tourist season is at its height, and being less frequent during the winter months.

Children under 11 years of age are admitted free when accompanied by an adult who will be responsible for their behavior. The admission for adults is \$1.50, including tax, and with the ticket in his hand the visitor waits for the next tour to begin.



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Sheet Stalactites—Queen's Chamber

As the guide collects the many sightseers together who will form his party, the enthusiasm for the trip within each individual mounts to a high tension. At his beckon the members of the party move toward the natural entrance—ready to begin an experience not quite like anything they have ever seen or known—an experience that will linger in their memories as long as they shall live.

A view of the Caverns is just ahead!

A park ranger briefs the party on the early history of the Caverns, describes its geological aspects, and reviews once again the rules visitors must observe on their trip below. Then, as he flicks a switch, the first section of the black hollow in the earth becomes illuminated. With a swish of his flashlight he directs the members of his party to follow.

The trip through the Caverns has begun!



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Elephant Ears—Queen's Chambers

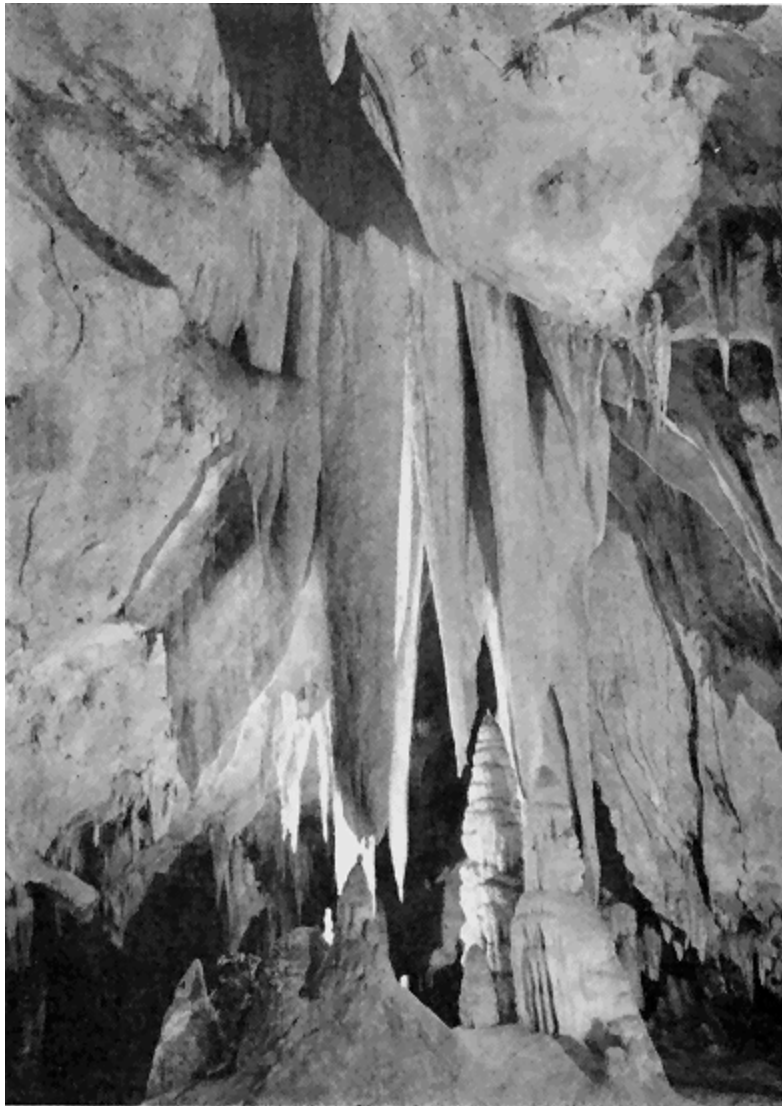
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From Below

Of several ways to enter the Caverns, only two are permitted today by the National Park Service. One is through the natural entrance and the other is by elevator which takes the visitor rapidly down 754 feet to a spot adjacent to the lunch room and the famous Big Room. The true adventurer—the visitor who wants to really see the Big Cave from the beginning, much as my father first saw it, will prefer to enter through the natural opening with the regular tour. Visitors not in the best of health will go down by elevator, but they will miss much of the fun and excitement.

As the tour passes into the mouth of the natural opening the warm sunshine soon gives way to the yellow incandescence of the man-made light which illuminates all the passageways and underground chambers. The cool cave air is soon felt, and the warm coat or sweater is already appreciated. The guide points out the important features that should not be missed, and answers questions from members of his party about the cave, its history, formations and development.

Visitors today have a sloping paved trail to follow—far easier than the rock strewn path the early explorers crawled over in their thirst for a view of what lay below. Steep climbs have been eliminated by sharp switchbacks which lengthen the walking time but make the descent much easier.



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Draperies—Papoose Room

As the tour continues, the descent begins to steepen and the walls change in both composition and texture. Solid rock predominates, appearing much darker than around the entrance. An occasional pale color is noticed in the illuminated areas, and a few moments later the first stalactite is seen hanging from the ceiling.

Shortly the tunnel divides and the party will follow the trail as it leads to the left. The guide points out that to the right is the Bat Cave, that portion of the Caverns where the millions of bats live and where the great guano deposits

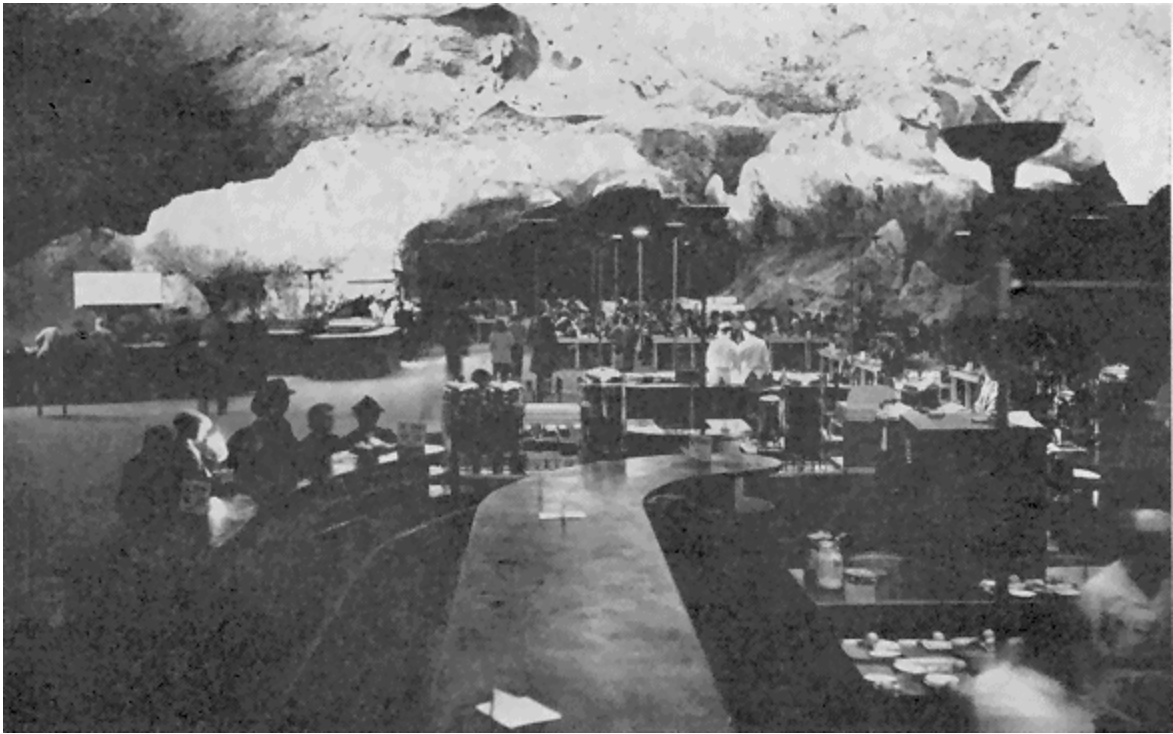
were found. It extends over a half mile back in an easterly direction and is not a part of the tour.

At this point the trail levels somewhat and the visitor finds himself in the Auditorium, a large room where the Ranger pauses for a brief explanation to the group. Members of the party will be surprised at how well the guide's low voice can be heard, a phenomenon due to the excellent acoustics of the room.

All eyes are absorbed in the ever-changing spectacle that is unfolding before them. Next comes a low ceilinged passageway, followed by a clear pool of water from which rises a lone column. The formations along the way are getting larger with each step, the coloration is more vivid, and the variations are great indeed.

Formations at times suggest the ancient cliff dwellings found at Mesa Verde, and one wonders if the early Indian did, by chance, ever inhabit them. Suddenly, as the descent drops rapidly, the ceiling is some 125 feet above and the visitor finds himself at the Whale's Mouth, a formation with slender stalactites enhanced by clever lighting which indeed does resemble the interior of that sea-going mammal's jaws.

As the ceiling rises still more, the party enters the Main Corridor, a tunnel almost a mile long which leads directly to the most picturesque rooms of the Caverns. The descent is for the most part uneventful and rapid, and the guide points out some of the unique formations along the route such as the American Eagle, which has a wingspread of 12 feet, the Baby Hippo, others resembling sea foam and waves, and the three Little Monkeys perched well towards the ceiling overhead.



The Lunchroom

Along the walls of the Main Corridor one may see small openings which reveal an inky blackness that may cover up some mysterious room or hallway. Someone in the party asks the guide about it. He explains that there are many such openings in other rooms of similar nature, but since they have nothing different to offer from what is already being seen, the Parks Department has not opened them to public view.

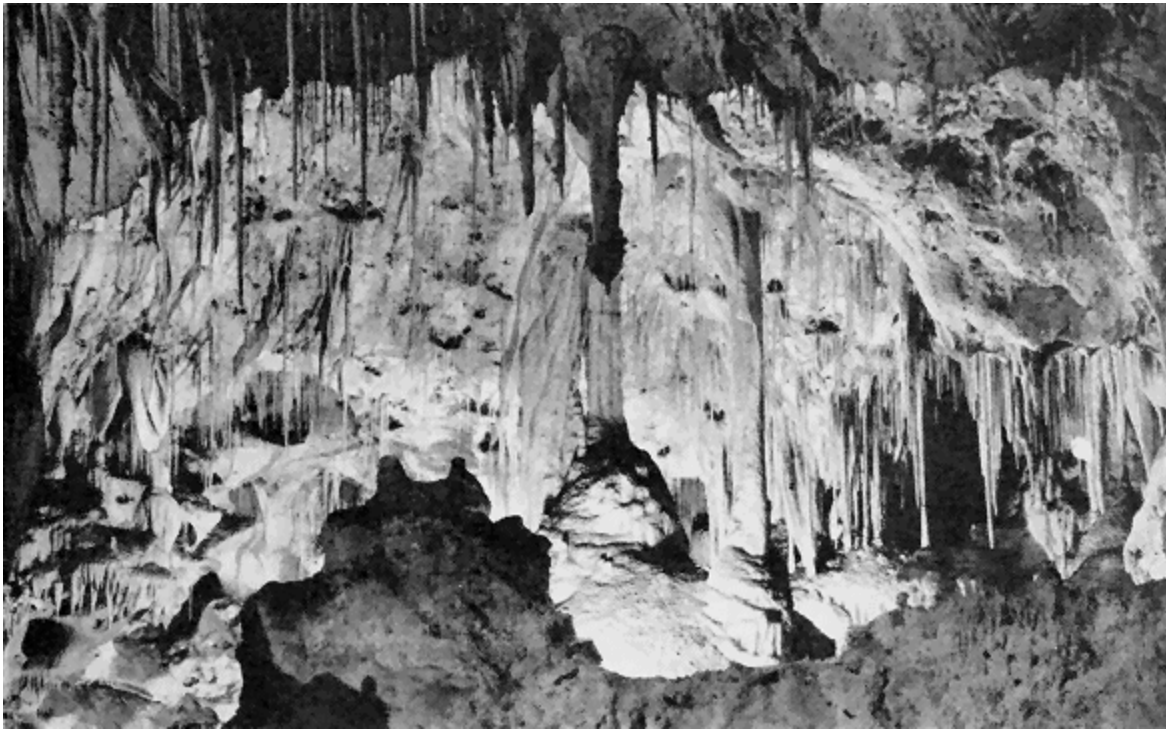
Have the Caverns been fully explored, someone asks. No, he replies. To date about 23 miles have been mapped and surveyed, and many openings into large chambers are known to exist, but their size is unknown. Perhaps even larger rooms lie close by. No one knows.

The formations continue to change in size, shape, and contour. Stalactites hanging from the ceiling look as though some little elf might have hung them there the night before, using a spindly stalagmite to stand on as he did so. The visitor has been so absorbed in what he has been seeing that he is startled to hear the guide announce that the party is now 600 feet below the surface. That is further than the Washington Monument is high.

The most picturesque portions are now to begin, he says, but the first of them will require some climbing. For that reason, those not quite up to the task may take a short cut to the lunchroom which is a brief jaunt to the left. The rest will head on past the Iceberg, the largest "loose" rock in the Caverns, which is estimated to weigh about 200,000 tons. It once "fell" from the ceiling, but the fall was gradual. The delicate stalactites grew later. The ceiling at this point is some 358 feet above.

If the visitor thinks he has seen something up to now, he is due for a surprise, for a few moments later the winding pathway leads to the first of the many scenic rooms for which the Carlsbad Caverns have become so famous. Appropriately named the Green Lake Room, this first of several similar chambers is breath-taking in its beauty. Its name is derived from the small green pond seen to the right of the trail, which is fed by drip water from above, a pool so clear its depth is easily seen.

On the ceiling above are huge masses of stalactites, at times so thick as to form curtains of stone, varying from reddish brown to a light grey, intriguing in their staggered form and defying description by even the most talented poet or writer. On beyond the pool the visitor marvels at the Veiled Statue, a pillar of dripstone which resembles a frozen waterfall and which began its existence about 100,000 years ago.



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The Painted Grotto—Big Room

Many visitors have said at this point they have had their money's worth. This one room alone was worth their whole trip. Actually, they aren't yet quite half way, and there is more to come.

Continuing on past the Frozen Waterfall and the Bashful Elephant, both named for the effect they give, the traveler enters the King's Palace which, to some, is the most ornate of the many chambers the entire Park has to offer. Thousands of stalactites hang from the ceiling, at times resembling stringy mosses hanging from the limbs of trees. Some, appearing to hang at a slight angle, give the impression the "stone moss" might be swaying in the wind. Underneath there is a noticeable scarcity of corresponding stalagmites, and geologists think the floor at this point was at one time covered with water, which would prohibit the formation of the "floor icicles."

The eye roams over ever-changing forms with no two quite alike, and the immensity of the room alone is sufficient to remind man what a small mortal he be, after all. At the far end of the Palace are seen two pendants,

stalactite and a stalagmite which come within a fraction of an inch of kissing. Now dead, they will never touch; hence the name, the Frustrated Lovers. And, as though they might have been the King and the Queen, they are found at the small passageway that connects the King's Palace with the Queen's Chamber, another room of similar shapes and decoration which do indeed remind the visitor of the ornate finery to be expected in Her Majesty's domicile. The countless rows of stalactites look like driplets of finely woven lace, stretching over much of the ceiling not unlike the frills of a canopy one might expect to find adorning the Queen's bed in her castle.

One unique and different formation is known as the Queen's Draperies, which are found at the southwest side of the room. They are a form of massed stalactites grown together so evenly that they resemble a huge drape from which the wrinkles haven't been ironed out. But the unusual feature is the ability of this stone to absorb light and reflect it moments after the light has been extinguished. These portieres are the highest in the Caverns, being slightly over 40 feet. Throughout the rest of the room one feels as though he must have interrupted Mother Nature while she was busily experimenting and trying to decide what to do next, having left behind her first efforts for the world to see.



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The Chapel or Dome Room in the Big Room

The paved trail continues through a low, rocky passageway at the far end of the Queen's Chamber and presently the party finds itself in the Papoose Room and in one of the lowest spots in the Caverns, approximately 875 feet below the surface. Here the stalactites have grown together in such a way as to resemble huge curtains of stone which appear much like a sheet of lava which has slipped through a crevice and then hardened.

Here, too, are numerous needle-like points protruding down from the ceiling as though sharp pointed pins had been pushed through from above. Areas on the floor remind the visitor of a growing rock garden, yet there is not a green leaf anywhere to be seen. Nevertheless, few gardens offer any more beauty or color than these gentle formations along the edges of the Papoose Room, smallest of the scenic chambers on the trail.

Here the party must retrace a few steps, returning through the King's Palace, leaving this time by the eastern end, entering a small passageway which leads to a series of sharp turns and switchbacks made necessary in order to climb the rather steep grade ahead.

It has now been an hour and 45 minutes since the visitor left the outer world behind, and the up and down climbing has done its part towards working up an appetite. But if that hasn't been enough, going up this zig-zagging hill should complete the task and make anyone ready for something to eat. Appropriately enough, it has been tagged Appetite Hill, and uses 60 feet to accomplish its task.

A moment later the party passes by a room of rocks which looks as though it was noon hour at a rock quarry, and the workmen had scurried off at the abrupt blow of the whistle, not caring in what manner of disarray they left their chores. But in this case it was Nature's workmen who left—untold hundreds of years ago—never to return. Labeled the Bone Yard, it features an alcove where the formation resembles a huge sponge—rock eaten away or dissolved by the continuous action of water which, at times, resembles bone formations.

After having seen the more colorful scenic rooms, many visitors give this spot scarcely more than a fleeting glance, but geologists find here exacting evidence as to the Caverns' beginning. After the Swiss cheese effect was created the water was in some manner diverted and, in the absence of drip water, no stalactites or stalagmites were formed.



© BY ROBERT NYMEYER

Altar in the Chapel—Big Room

A few steps more and the visitor finds himself in a huge room with a massive rock formation hanging from the ceiling as though it might fall to the floor at some prearranged signal. Huge boulders dot the sides of the room, between which are many narrow white benches and tables. In the center of the room are long aisles which carry the visitor past waist-high counters on which have been placed hundreds of box lunches. Yes, this is the famous lunchroom, 754 feet below the surface of the earth, and just about the most popular room for those who have been hiking for the past couple of hours.

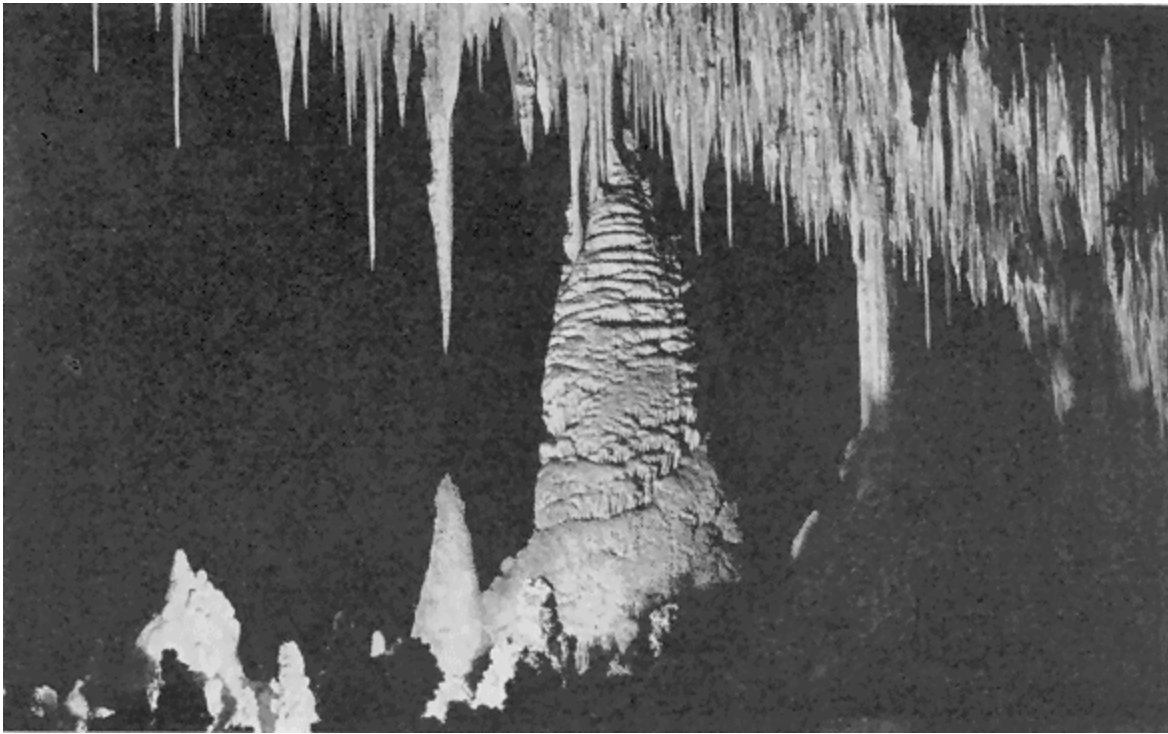
The guide announces there will be a rest of 40 minutes—ample time to eat a noon-day meal and rest a couple of tired feet. It seems that no matter how many people enter the lunchroom, the line past the counters never stops or is held up. Designed to accommodate 1200 an hour, as many as 2000 hungry sightseers have been clocked picking up their lunch and beverage in that time. The lunchroom is operated by the Cavern Supply Company under the supervision of the National Park Service.

Visitors who took the easy way into the Caverns, via the elevator, arrive at the lunchroom as their first stop. From this point they continue their limited tour of the Caverns pretty much on one level.

With so many people in this one huge room a long way from the outside atmosphere, someone asks the guide if the air doesn't become stale and musty. No, he replies, the air changes naturally every 24 hours. How much longer does our tour last, another wants to know. This is the halfway point, says the guide, and we have about two hours more.

Here many a traveler pauses a moment to absorb what he has seen—the beauty, the magnificence, the grace and charm. Mother Nature does her work so well—so thoroughly. But the guide said there is more. How can there be more—how can anything match what has already been seen? Man is indeed a humble creature, the tiniest detail in the great heavenly scheme. Great though man's work may be, God's work is ever greater.

The rest period is over. It is time to go on. In a moment the party is once again assembled in the main passageway section which connects the many rooms, and turns to the left past "hanging mosses" and honeycombed fissures into the famous Big Room, largest underground chamber known anywhere, and unsurpassed in length, width and height, variety and size of formations, number of colors to be seen, and value to geologists due to the many finds that have been made.



© BY ROBERT NYMEYER

Temple of the Sun—Big Room

Here the remainder of the 4-hour tour will be spent. The trail around the perimeter of the room is one and a quarter miles and, although many sections and alcoves are actually under one huge roof, the visitor usually feels he is in a series of separate chambers. One reason is that the big room is shaped roughly in the form of a cross, the length of the "staff" measuring some 2,000 feet while the "cross-arm" is slightly over half that length. 2,000 feet is almost $\frac{4}{10}$ of a mile!

Two hours is not enough to see everything the big room has to offer. Many small formations are not seen but must be discovered. Every visitor sees something new—something no one else within the party has noticed, for the formations are countless and the resemblances they offer are equally as numerous.

The elaborate indirect lighting system has been designed to reveal the largest, most beautiful and ornate of the spectacles to be seen, but a flashlight aimed in a dark corner is likely to reveal formations resembling a Mermaid, Frog on a Toadstool, Queen's Necklace, Weeping Willow Trees,

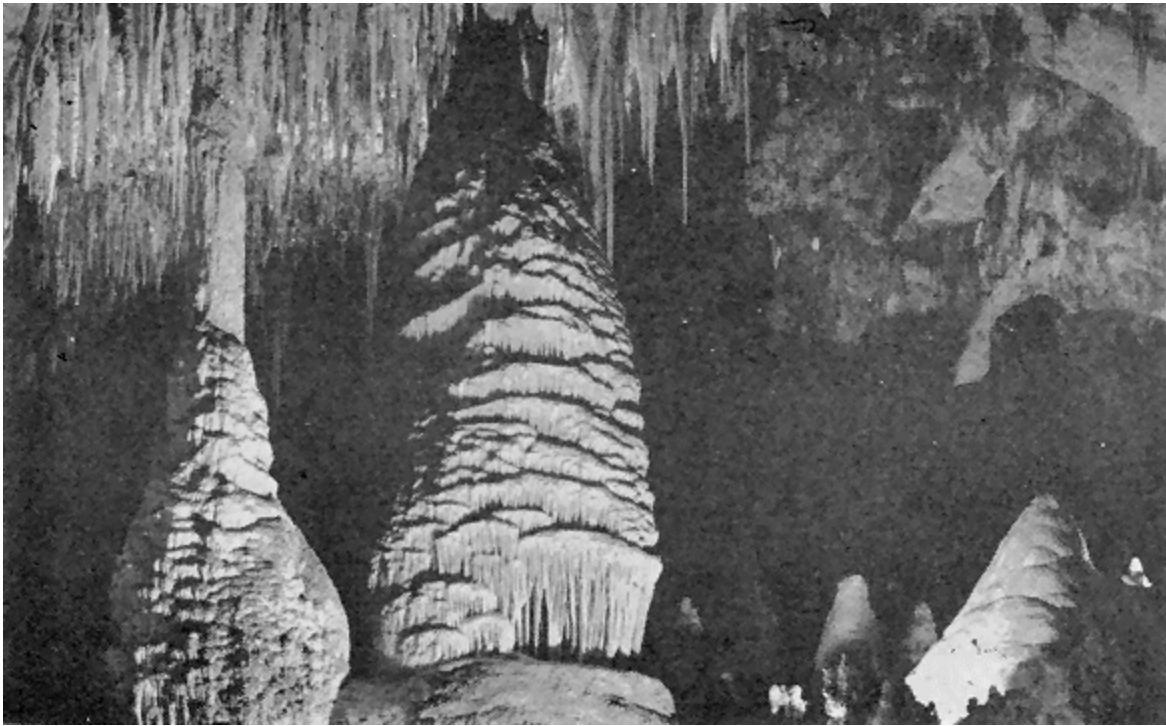
the Petrified Forest, and countless others, limited only by man's imagination and his ability to seek them out.

But the larger formations which have contributed to the fame of the Big Room include Giant Dome, Hall of Giants, Rock of Ages, and the Temple of the Sun.

No two people see quite the same thing when they enter the Big Room. It is so huge and there is so much to greet the eye that at first everything is taken in but no one thing is seen. Then the eye begins to settle on one formation at a time. One of the first is the Hall of Giants, so named for the size of its principal formations; the huge Giant Dome, a massive, light grey stalagmite flecked with tinges of orange, stretching upward some 62 feet above the ground, making it the largest upward growth in the Caverns, and behind it the Twin Domes, similar in structure and formation, though not as high and therefore not as old.

As the party continues one listens to the whispered observations of the members. Looks like the inside of a great Cathedral, says one. Oh, no, answers his companion, too eerie and spooky for that!

The beauty continues to defy all description. The color in the Big Room, too, is appealing to the eye, no doubt enhanced by the deft use of colored lighting. No, says the guide. Only plain light bulbs are used in the thousands of feet of wire strung throughout the Caverns. Any color you see was put there by Mother Nature herself.



© BY ROBERT NYMEYER

Temple of the Sun—Big Room

Creeping along the north side of the Big Room the party passes by Fairyland, a group of fantastically shaped stalagmites recalling the fairies from childhood tales as they go about their sprightly tasks.

Next to be seen is the Temple of the Sun, named for the many colors found shimmering on its sides and base. It is a huge stalagmite from which hang many stone icicles from a dozen or more layers, the curtains of icicles pink in color mounted upon a stone base of blue.

Thousands of inverted spires rain down from above, and on the ground the crinkly rock formations resemble pop-corn in color and appearance although not hardly in flavor! This same pop-corn formation is seen in other parts of the Caverns and geologists say this condition resulted after stalagmites had grown to a considerable height, following which the cave was flooded for thousands of years with water, the excess lime from which accumulated in the tiny globules which give the bubbly appearance.

Further on is the Totem Pole, a tall, slender stalagmite whose parent stalactite never had time to form due to the rapid rate of the dripping water.

Standing pretty much alone, it sticks out like a large suhuaro cactus stalk against a cloudless sky. The outside "skin" is about as rough, but the spines are missing. To others it resembles a group of men's hats thrown carelessly into a pile, one on top of the next. What it lacks in color, it makes up for in its unique appearance.

The floor of the Big Room is a reddish brown color, and at this point the visitor sees numerous rounded mounds of stones, looking much as though a stalagmite might have gotten a good start only to be engulfed in water and be polished down to this semi-globular formation. That is probably just what happened, leaving these half-buried basketballs to remain in exactly that fashion for eternity.

A sudden right turn brings the party to the famous Jumping Off Place, which in reality is one of the entrances to the Lower Cave. The trail passes midway between floor and ceiling, approximately 150 feet in each perpendicular direction. Here the floor of the Lower Cave approximates 900 feet below the surface.

An easier entrance provides access to the cave which may be entered by those who care to do so, although it is not a part of the regular tour. Lower Cave shares honors with the Papoose Room as being the two lowest points on the underground tour. Also, they are the furthest west. Here, with the limit reached, the pathway begins its return trip.



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Giant Column and Twin Domes—Big Room

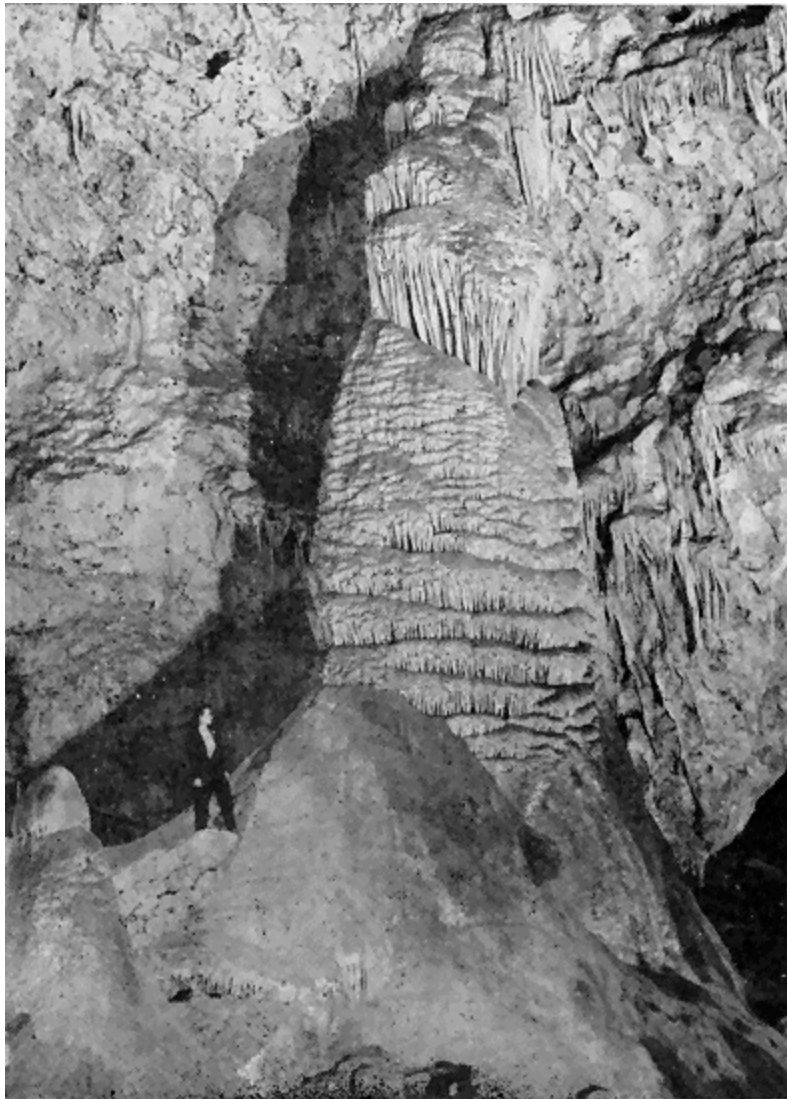
On once more, lily pads, huge boulders, and clusters of flowstone along the walls greet the visitor as he heads for Mirror Lake, a small, crystal-clear pool that is well named. On beyond is the Bottomless Pit, a dark hole with mysterious implications that received its name from the first explorers who thought the title would intrigue all who saw it. From the trail the light of a spotlight has trouble in penetrating the depths of the pit, adding to its ghostliness. Actually, the pit has a bottom some 140 feet below the rim where the visitors pass, but unlike many other attractions of the Caverns which were named by those who first saw them, the name of the Bottomless Pit has remained unchanged. Other of the Caverns' features have been renamed two or three times within the past few decades.

As the column of silent visitors marches silently along the return trail the Big Room takes on a new look. Indeed it does not appear to be the same room where the party entered slightly over an hour ago. All sense of time and direction are lost in the myriad of decoration, light and shadow, gigantic, often grotesque formations, and the serenity of this underground heaven.

Passing quietly ahead the traveler pauses at the largest "living" stalagmite within the known part of the Caverns, the Crystal Spring Dome. Generally dry, this part of the Big Cave is moist, and water, dripping slowly from the ceiling above, is still depositing its minute quantities of calcium on the mighty Crystal Spring Dome as it has been doing for countless centuries.

The Crystal Spring is perhaps 20 feet high and resembles a bushy Christmas tree laden with snow, its branches drooping under the massive white blanket. The dripping water is depositing its lime at the rate of about 2½ cubic inches a year, although this rate varies depending upon the amount of moisture descending from the ceiling above, and also upon the amount of lime contained in the water.

This constant application of moisture keeps the white, plump stalagmite glistening all the time, and the excess moisture, unable to evaporate, forms into a small pool at the base of the dome, known as Crystal Spring.



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Rock of Ages—Big Room

In an adjoining area is the celebrated Rock of Ages, a gigantic stalagmite, stately in position and dramatic in location, thought to be one of the oldest forms in the Caverns, although its exact age is not known. For years it marked the end of the Caverns Tour, and an impressive ceremony conducted by the guides consisted of a short talk followed by the singing of the old favorite, "Rock of Ages." No visitor could ever forget this inspirational ending to a breath-taking tour of God's underground handiwork. The Rock of Ages became almost a symbol of the Caverns tour

in the minds of those who had taken it. But, as the crowds of visitors increased, the park officers found they could not continue this service.

Its discontinuance brought protests from former Caverns visitors who had remembered this final closing gesture as the high point of the tour. Rippled layers give the Rock of Ages a shape all its own, and it marks the end of the journey through the mighty Carlsbad Caverns of today.

It is now nearly four hours since the traveler entered the big natural opening to the cave some 700 feet above him. His sweater feels good, though he isn't as cold as he had first supposed he might be, for the constant walking during the 2¾ miles which the full trip requires has kept him relatively warm.

Before saying good-bye to the Caverns he likes to pause a moment at the entrance to the Big Room and look back, for one last glimpse of this largest known room in the world, man-made or otherwise. Above him some 285 feet is the bedecked ceiling, the highest point in the Caverns, and suspended by the longest unsupported arch known in the world!

The human mind can't quite believe it, yet there it is. And as the visitor slowly turns to thoughts of the return trip he revolves in his mind the beauties he has seen.

He will find the elevator takes him quickly and safely 750 feet upward to the surface where the warm sunlight appears exceedingly bright to human eyes, which, in the intervening hours, have become accustomed to the gentle illumination found in the Caverns' otherwise inky blackness.

Within his own mind the visitor knows he will never forget the indescribable sights he has seen, but how can he tell others of the size, the intricate design, the scope and beauty that he has witnessed? Pictures help, and these may be purchased at the souvenir stand near the entrance. Views of all the better known formations in color are obtainable, as well as booklets telling about the history of the mammoth Caverns and how they were formed millions of years ago.

Man enters the great underground fairyland a bit apprehensive and sometimes skeptical as to what he will see, finding it hard to believe all he has heard of the Great Carlsbad Caverns of New Mexico. But some four hours later, as he once again greets the light of day, he recollects how Man

has been responsible for some pretty remarkable shows, it is true, but now he is convinced that Mother Nature herself puts on the biggest, the most colorful, and the most spectacular show of all!

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The famous bat flight—a nightly occurrence at the Caverns.

Transcriber Note

Some text was moved to avoid paragraphs being split by illustrations. Two images in the original volume were negative images of documents. These have been replaced with positive images with links to the original images. Transcriptions for images of printed documents are provided to assist the reader. As a handwritten and typeset version of Amended Notice of "Big Cave" Mining Claim were presented and labelled the same, "(Handwritten)" was added to the caption of the first image to distinguish them. Several transcribed copies contain typos as given in the original.

*** END OF THE PROJECT GUTENBERG EBOOK THE BIG CAVE

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 day located the following described Placer Mining Ground containing Twenty (20)
 Acres to wit: from Monument 300 ft to where E 1000 ft corner the
 claim taken up for Luans or Silver Soil Copper or any other Mineral about
 25 miles South West of Carlsbad N. M. and near Walnut Canyon and about
 1 1/2 miles South East of the Donation Spring situated in Eddy Co. N. M. contain-
 ing valuable deposits of Luans clay and other kindred substances situated in Eddy
 Co. Mining District Grant, of Eddy County of New Mexico.
 This record sheet is given as the Big Horn Mining claim.
 Located at the 25th day of Nov. 1893.
 Attest A. L. Furschfeld
 J. H. Long

1

Abijah Long's original claim to the Carlsbad Caverns

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PHIL. D. SWING
17TH DIST. CALIFORNIA

COMMITTEE:
IRRIGATION AND RECLAMATION
PUBLIC LANDS
FLOOD CONTROL
EXPENDITURES IN THE EXECUTIVE
DEPARTMENTS

Congress of the United States
House of Representatives
Washington, D. C.

December 24, 1930.

Mr. H. F. Shepherd,
Deputy in charge,
United States Customs Service,
Calxico, California.

My dear Mr. Shepherd:

This will acknowledge your letter regarding Mr. Abijah Long who is the original discoverer of the famous Carlsbad Cavern in New Mexico.

Sincerely,

Phil D. Swing

PDS:DCM

RECEIVED
CUSTOM HOUSE
DEC 24 1930

Letter from U. S. Congress recognizes Long as cave's discoverer.

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